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Statistical static timing analysis: A survey

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ABSTRACT

As the device and interconnect physical dimensions decrease steadily in modern nanometer silicon technologies, the ability to control the process and environmental variations is becoming more and more difficult. As a consequence, variability is a dominant factor in the design of complex system-on-chip (SoC) circuits. A solution to the problem of accurately evaluating the design performance with variability is statistical static timing analysis (SSTA). Starting from the probability distributions of the process parameters, SSTA allows to accurately estimating the probability distribution of the circuit performance in a single timing analysis run. An excellent survey on SSTA was recently published [D. Blaauw, K. Chopra, A. Srivastava, L. Scheffer, Statistical timing analysis: from basic principles to state of the art, IEEE Trans. Computer-Aided Design 27 (2008) 589–607], where the authors presented a general overview of the subject and provided a comprehensive list of references.

The purpose of this survey is complementary with respect to Blaauw et al. (2008), and presents the reader a detailed description of the main sources of process variation, as well as a more in-depth review and analysis of the most important algorithms and techniques proposed in the literature that have been applied for an accurate and efficient statistical timing analysis.

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1. Introduction

As microelectronic technology continues to reduce the minimum feature size, and consequently to increase the number of transistors that can be integrated onto the same die in accordance with the Moore's law, the gap between the designed layout and what is really fabricated on silicon is widening significantly. As a consequence, performances predicted at the design level may drastically differ from the results obtained after silicon manufacturing. Aggressive technology scaling introduces new sources of variation, while at the same time process control and tuning during fabrication become more and more difficult. Coping with variations during design has potentially significant advantages both in terms of *time-to-market* and reduced costs in process control. The first ones stem from taking the right decisions early in the design flow, even at the system level, thus considerably reducing the number of design iterations before tape-out. Furthermore, variability reduction by means of process control usually requires expensive manufacturing equipment [1]. Hence, the impact of parameter variations should be compensated with novel design solutions and tools, due to the very high cost of advanced process control techniques [2,3]. Following the technology scaling, while steadily shrinking in absolute terms, process variations are growing as a percentage of increasingly smaller geometries [4,5]. Moreover, variability sources grow in number as the process becomes more complex, and correlations between different sources of variation and a general quality figure of the process are becoming more and more difficult to predict. Manufacturing variations introduce the following yield loss mechanisms:

- *Catastrophic yield loss*: Fabricated chips do not function correctly.
- *Parametric yield loss*: Fabricated chips do not perform according to specification (they may not be as fast as predicted during design, or may consume more power than expected). In designs that are at-speed tested and binned in conformity with their performance like microprocessors, dies are targeted to different applications in line with their performance level, and parametric degradation means that fewer chips end up in the high-performance, high-profit bin. In other design styles like ASICs, circuits below a performance threshold must be thrown away.

Obviously, the catastrophic yield loss has traditionally received more attention. Typical functional failures are caused by the deposition of excess metal linking wires that were not supposed to be connected (bridging faults), or by the non-deposition of metal thus leading to opens. Techniques to handle catastrophic yield loss include critical area minimization, redundant via-insertion, wire widening/spacing, and methods like design centering and design for manufacturing (DFM). In contrast, the parametric yield loss is becoming more and more important since design performances are dramatically affected by process variations, as illustrated in Fig. 1. For designs based exclusively on optimization of the nominal process parameters, the analysis may be inaccurate, and synthesis may lead to wrong decisions when the parameters deviate significantly from their nominal value. For a long time parametric yield loss has been an overlooked problem. Recently, a strong research effort has been devoted to this topic, and this survey is focused on parametric yield loss.

Typically, the methodology to determine the circuit timing performance spread under variability is to run multiple static timing analyses (STA) at different process conditions, i.e., "cases" or "corners", which include the "best-", "nominal-" and "worst-case". A process corner (or corner in short) is a set of values assigned to all process parameters to bound the circuit performance. The worst-case corner is defined as the corner with every parameter at the $\mu \pm 3\sigma$ value, such that a typical circuit has the smallest slack. However, it is worth pointing out that determining the real worst-case corner is very difficult (if not impossible at all) without an explicit enumeration of all corners, since the circuit slack is a non-monotonic function of variation parameters. This approach is breaking down because the increasing number of independent sources of variation would require too many timing analyses. In fact, the corner-case approach necessitates up to 2^n runs, where n is the number of significant sources of variation. In Table 1, a list of the principal variability sources in advanced

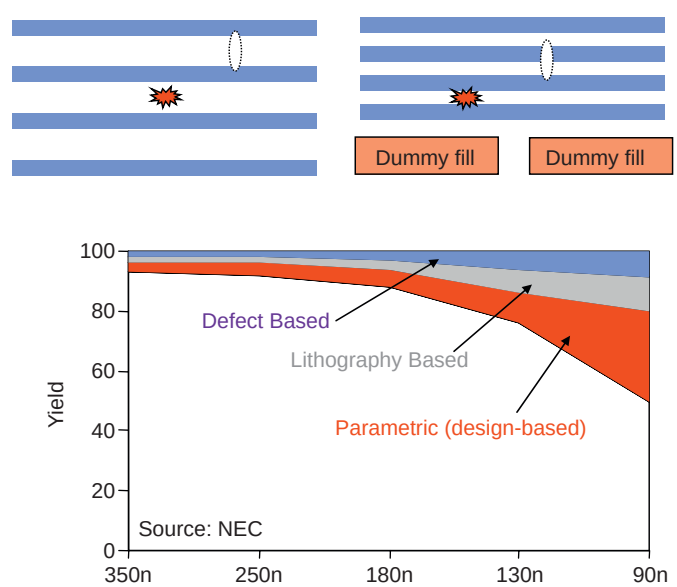

Fig. 1. Catastrophic vs. parametric yield loss.

Table 1
Variation impact on delay (*Source*: L. Stok, IBM [6]).

Parameter	Delay impact
BEOL metal (metal mistrack, thin/thick wires)	−10% to +25%
Environmental (voltage islands, IR drop, temperature)	±15%
Device fatigue (NBTI, hot electron effects)	±10%
V_{th} and T_{ox} device family tracking (can have multiple V_{th} and T_{ox} device families)	±5%
Model/hardware uncertainty (per cell type)	±5%
N/P mistrack (fast rise/slow fall, fast fall/slow rise)	±10%
PLL (jitter, duty cycle, phase error)	±10%

silicon technologies and their impact on delay is reported [6], and a complete case analysis taking into account all these variations may need from 2^7 up to 2^{20} timing analyses! A possible solution to reduce the number of timing analyses is to design and verify in the worst/best-case corner. Worst/best-case timing analysis determines the chip performance by assuming that worst/best process and operating conditions exist simultaneously. Therefore, the delay of each circuit element is computed under these conditions. Since only the performance extreme values are of interest, neither the details of the performance probability density function (PDF), nor the distribution of the single parameters are necessary. This approach is based on the assumption that if a circuit works correctly under the most pessimistic conditions, then it will function under nominal conditions. Hence, designing in worst-/best-case would automatically take into account the nominal case. However, considering the corner values for each electrical parameter may lead to over-pessimistic performance estimation, since the actual correlation between electrical parameters is not considered. In other words, the scenario with all parameters in their worst-/best-case values has really a minimal probability to happen in practice, and in several cases it cannot happen at all. As an example, by considering the variation impact on delay reported in Table 1, the worst-case approach will give a [−65%, +80%] guard-band timing interval, thus leading to a strong underutilization of the technology. Furthermore, within-die (WID) variations have become a non-negligible component of the total variations [4,5]. These variations may be handled by existing corner-case design methodology only by applying different derating factors for datapath and clock-path delay, and/or by introducing large uncertainty margins, resulting in either an over- or under-estimation of the circuit delay, depending on the circuit topology. Another drawback of the traditional worst-case methodology is that it cannot provide information about the design sensitivity to different process parameters, which could potentially be very useful to obtain a more robust design implementation. Examples of worst-case approaches can be found in [7,8].

A potential solution to the problem of accurately evaluating the design performance with variability is statistical static timing analysis (SSTA). Starting from the probability distributions of the sources of variation, SSTA allows to computing the probability distribution of the design slack in a single analysis. An example of the design slack distribution is illustrated in Fig. 2. The plot indicates that for a slack of −200 ps the parametric yield of the design will be close to 100%, while for a slack of 300 ps the yield drops to about 0%. The slack distribution information may yield several advantages. For products that are at-speed tested and binned like microprocessors, it allows to predicting the number of chips that will fall into the high-frequency bin, and consequently sold for the highest profit. More in general, it allows to estimating the true operating frequency. In contrast, for ASICs, it permits an early decision making on risk management at chip level. Another important output from SSTA is diagnostics, enabling a designer or an automatic optimization tool to improve the circuit overall performance and robustness, by exploiting the sensitivity of the arrival times to different sources of variation. Therefore, SSTA will simultaneously allow to targeting high-performance while providing quantitative risk management [9].

This survey is organized as follows: in Section 2 the most important sources of device and interconnect variations are introduced and classified. In Section 3, the formulation of the SSTA problem, the key challenges, and the different approaches are presented, while the main algorithms and techniques adopted in modern block-based SSTA are described in Section 4. Finally, Section 5 presents some conclusive remarks.

2. Sources of variation

Process variations in both interconnect and devices dictate more conservative design margins. Therefore, understanding how much variability exists in a given design and its impact on timing and power performances is becoming a critical issue. In the following sections, the impact of different variability sources is analyzed.

2.1. Definition and classification

Variation is the deviation from designed values for a layout structure or circuit parameter. The electrical performance of VLSI ICs is impaired by two principal sources of variation:

- *Environmental variations*, which arise during the circuit operation, and include fluctuations in power supply, switching activity, and die temperature. These variations are time-dependent and have a large range of temporal time constants that vary from the nanosecond to millisecond for temperature effects. Therefore, they are also called temporal (or dynamic) variations, and directly impact the parametric yield.
- *Physical variations*, which arise during manufacturing and result in structural device and interconnect parameter fluctuations. They include lithography-induced systematic and random variations in critical device dimensions such as transistor length and width, as well as wire and via width. Moreover, they also include random phenomena like the impact of discrete dopant fluctuations on MOSFET threshold voltage, and systematic phenomena like inter-layer dielectric thickness

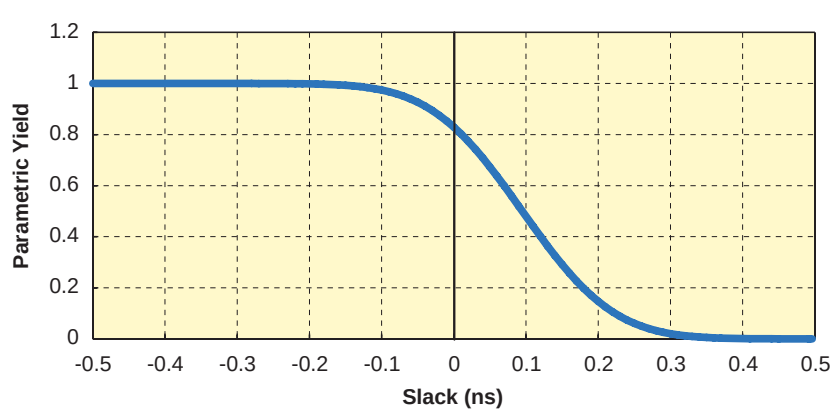

Fig. 2. Design slack distribution.

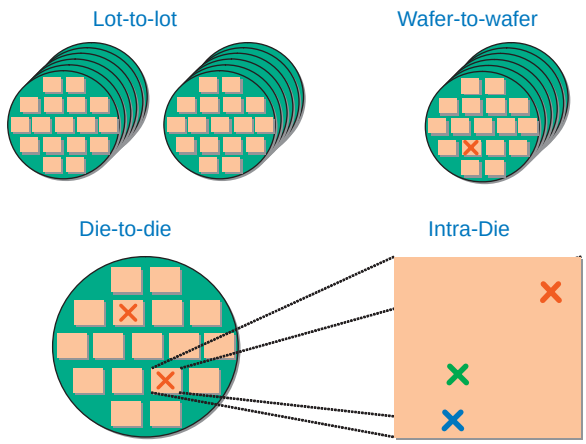

Fig. 3. Classification of physical variations.

variations with layout density (due to chemical–mechanical planarization). Such variations are essentially permanent; they are also called spatial variations, and may reduce the parametric yield, and potentially introduce catastrophic yield loss.

It is important to note that both environmental and physical variations depend on the design implementation. For example, device size variations due to lithography are a strong local function of layout, while power supply fluctuations are clearly dependent upon placement and power distribution network design. This has deep implications on the applicability of SSTA in the context of a realistic design flow. Physical variations can be further decomposed into different contributions, including *lot-to-lot*, *wafer-to-wafer*, *within-wafer*, and *intra-die* (also known as *within-die* or *on-chip variation*, i.e., *OCV*), as summarized in Fig. 3. Basically, for circuit design, physical variations might be simply separated into *inter-die* and *intra-die* components. Recently, the intra-die variations have become a real concern to the performance and functionality of complex digital ICs [10,11], since after poly-gate length (i.e., device critical dimension) has decreased below the wavelength used in optical lithography, both the systematic and random intra-die channel length fluctuations have exceeded the die-to-die deviations [12].

2.1.1. Inter-die variations

Inter-die variation is the difference of some parameter values across nominally identical dies (where those dies are either fabricated on the same wafer, or different wafers, or come from different lots), and in circuit design is typically modeled with the same deviation with respect to the mean of such parameters (i.e., threshold voltage V_{th}, or wire width on a given metal layer W_{intl}) across all devices or structures on any chip. It is assumed that each contribution in the inter-die variation is due to different physical and independent sources, and it is usually sufficient to lump these contributions into a single effective die-to-die variation component with a unique mean and variance. For example, the transistor channel length distribution can be obtained by silicon measurements from a large number of randomly selected devices across chips on the same wafer (or different wafers and lots); then, the mean and variance are estimated from the approximately normal distribution of these devices. In this straightforward approach, called the "lumped statistics", the details of the physical sources of these variations are not considered; rather, the combined set of underlying deterministic as well as random contributions are simply lumped into a combined "random" statistical description.

2.1.2. Intra-die variations

Intra-die (or WID) variation is the parameter spatial deviation within a single die. Such WID variation may have several sources depending on the physics of the manufacturing steps. For inter-die variation equally affecting all structures across several dies, the concern is how a variation that "rises or falls" in unison across the die may impact on performance or parametric yield. Moreover, the intra-die variation contributes to the loss of matched behavior between structures on the same chip, where individual MOS transistors, or segments of signal lines, may vary differently from designed or nominal values, or may differ unintentionally from each other. Two sources of WID variations are particularly important:

- *Wafer-level variations*, whose effects are small fluctuations across the spatial range of the die. As an example, many deposition steps might introduce systematic variations across the die.
- *Layout dependencies*, which may create additional variations that are increasingly problematic in IC fabrication. As an example, two interconnect lines identically designed in different regions of the die may have different widths, due to photolithographic interactions, plasma etch micro-loading, or other causes. Distortions in lens and other elements of the lithographic system also create systematic variations across the die. The range of such perturbations can vary: line distortions in exposure is within the range of a micron or less, while film thickness variations arising in chemical–mechanical polishing (CMP) may occur in the millimeter range.

While such variations may be systematic in any given die, the set of these variations across different dies may have a random distribution. For this and other reasons (i.e., lack of layout information), systematic variations are often bounded by, or treated as, some large estimated random variations. The physical process variations can be further categorized depending on whether they impact device, or interconnect characteristics.

2.1.3. Device variations

The active device variations, also denoted as Front-End-of-the-Line (FEOL) variations, include:

- *Lateral dimension (length, width) variations*, which are typically due to photolithography proximity effects (systematic pattern dependency), masks, lens, or photo system deviations, and plasma etch dependencies. MOSFETs are well-known to be particularly sensitive to effective channel length L_{gate}, (and thus to poly gate length), as well as gate oxide thickness T_{ox}, and to some degree also to the channel width W_{gate}. Channel length variation is often singled out for particular attention due to its direct impact on device output characteristics.
- *Doping variations*, which are due to implant dose, energy, or angle variations, and can affect junction depth and dopant profiles (and thus also impacting the effective channel length), as well as other electrical parameters such as threshold voltage V_{th}. Another source of V_{th} variation is related to random dopant fluctuations due to discrete location of dopant atoms in the channel and source/drain regions [13].
- *Deposition and annealing variations*, which may result in wafer-to-wafer and within-wafer deviations, and may also have large random device-to-device components. These material parameter deviations can contribute to appreciable contact and line resistance fluctuations.

All these variations change the device properties and impact the circuit performance.

2.1.4. Interconnect variations

The interconnect variations, also denoted as Back-End-of-the-Line (BEOL) variations, consist of the following components:

- *Metal thickness T variations*, due to deposition deviations in conventional metal interconnects, or dishing and erosion fluctuations in damascene (i.e., copper polishing) processes.
- *Dielectric thickness H or ILD variations*, caused by fluctuations of deposited or polished oxide films. Furthermore, the CMP process can introduce strong ILD variations across the chip.
- *Line width W and line space S variations*, due to photolithography and etch dependencies. At the smallest dimensions (lower metal levels), proximity and photolithographic effects may be important, while at higher levels etch effects depending on line width and local layout can be more significant.
- *Line edge roughness (LER)*, due to the photolithographic and etching steps.

BEOL variations change the wire electrical properties, including resistance, capacitance, and inductance. These electrical parameters directly affect the circuit performance. The critical paths often contain long wires, and a good description of the interconnect geometry variation is necessary for accurate circuit timing analysis. It is important to note that the interconnect sources of variability are relatively uncorrelated to device variations; hence, the number of significant and independent variations can be very large. To summarize, the most important sources of variation in 90 nm (and below) CMOS technology are listed in Table 2, where for each component the classification as inter-die systematic, intra-die systematic, random, or as a combination of these are reported [14].

2.2. Variation trends

The works described in [4,5] considered the trends of process-induced variations and proposed a modeling and simulation technique to deal with this variability. They used a simple circuit composed of a buffer driving an identical buffer through the length of a minimum-width wire, and performed a simulation study of the circuit for five different technologies, from 250 to 70 nm gate-length range as defined in the 1997 SIA technology roadmap. The technology parameters and their 3σ variations are summarized in Table 3, where it is reported that the manufacturing variations are increasing relatively to their nominal values, as illustrated in Fig. 4. Furthermore, the intra-die variations are also increasing significantly, as shown in Fig. 5, which reports the ratio between WID and total variations for some key device and interconnect parameters. Following the technology scaling trends, CMOS devices are expected to continue shrinking over the next two decades, but as they approach the dimensions of the silicon lattice, they can no longer be described, designed, modeled, or interpreted as continuous semiconductor devices. Fig. 6 illustrates a 22 nm (physical gate length) MOSFET expected in mass production before 2010 according to the 2003 ITRS roadmap [15], where there may be less than 50 Si atoms along the channel. In these devices, random discrete dopants, atomic-scale interface roughness, and line-edge roughness will introduce large intrinsic

Table 2
Variation components in 90 nm CMOS technology.

Component	Form of variation
Channel length	Inter-die systematic, intra-die systematic, intra-die random
Threshold voltage	Inter-die systematic, intra-die random
Mean metal R and C differences between layers	Inter-die systematic
Voltage and temperature	Intra-die systematic
NBTI, hot electron	Intra-die systematic

Table 3
Technology process parameter (nominal/3σ variations) trends.

Year	L_{eff} (nm)	T_{ox} (nm)	V_{th} (mV)	W (μm)	H (μm)	ρ (mΩ)
1997	250/80	5.0/0.40	500/50	0.80/0.20	1.2/0.30	45/10
1999	180/60	4.5/0.36	450/45	0.65/0.17	1.0/0.30	50/12
2002	120/45	4.0/0.39	400/40	0.50/0.14	0.9/0.27	55/15
2005	100/40	3.5/0.42	350/40	0.40/0.12	0.8/0.27	60/19
2006	70/33	3.0/0.48	300/40	0.30/0.10	0.7/0.25	75/25

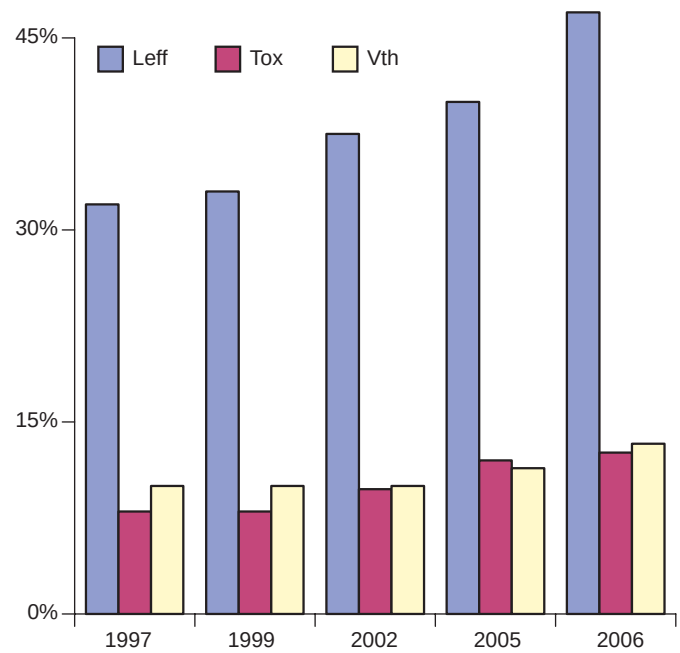

Fig. 4. 3σ parameter total variation vs. nominal value.

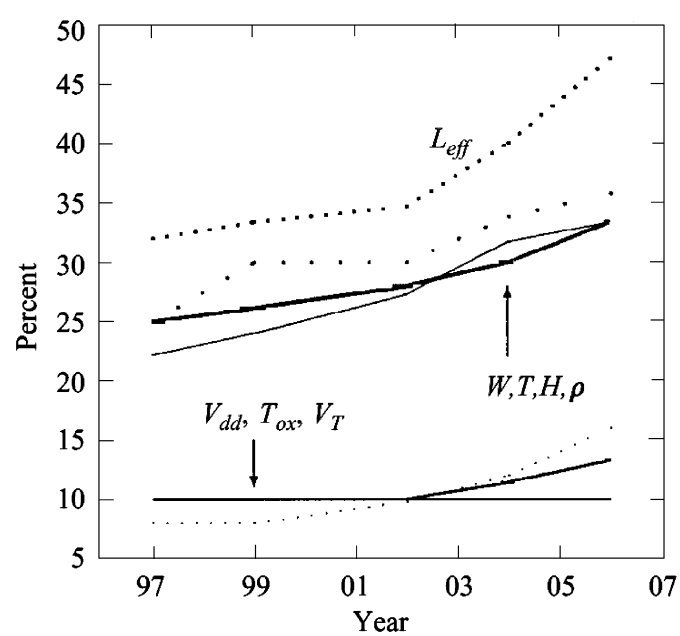

Fig. 5. Total variation percentage accounted for by intra-die variations [5].

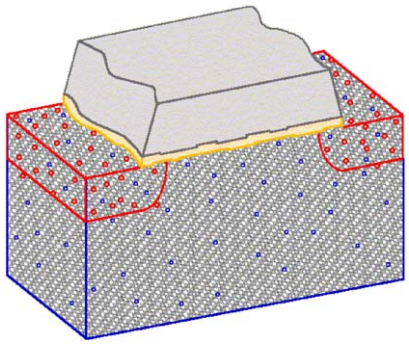

Fig. 6. A 22 nm MOSFET device.

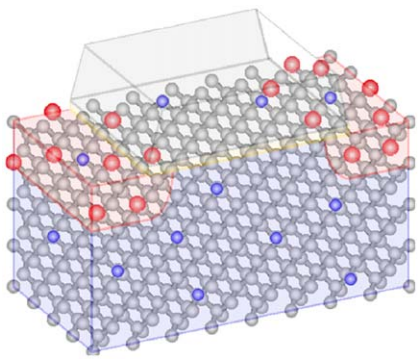

Fig. 7. A 4 nm MOSFET device.

parameter fluctuations. Fig. 7 sketches a 4 nm MOSFET predicted in mass production in 2020, according to the IBM roadmap, where less than 10 Si atoms are expected along the channel. Figs. 6 and 7 obtained from device/structure simulations show that MOS transistors are rapidly becoming truly atomistic devices and the random variations are becoming dominant.

3. Introduction to statistical static timing analysis

In traditional digital design, variations have been considered in the manufacturing process by guard-banding, using a corner-based approach. This method identifies "parameter corners", such that the 3σ-deviation of all manufactured circuits will not exceed these corner values, assuming that variations exist between different dies, but within each die the individual components such as transistors have the same behavior. However, this paradigm is breaking down. Random and systematic defects, as well as parametric variations, have a large detrimental influence on performance and yield of the designed and fabricated circuits. Manufacturing variations are increasing with respect to their nominal values, and new process technologies achieve much less benefit regarding performance and power consumption because of extensive guard-banding. Hence, guard-banding based on 3σ corners may soon become no longer economically viable. At the same time, as it was pointed out in Section 2, WID variations cannot be handled with the existing corner-based techniques. Currently, designers deal with these effects by including in traditional corner-case STA either the on-chip variation (OCV) derating factor, or by increasing the number of process corners. However, this approach does not capture the statistical nature of OCVs, and technology scaling has further exacerbated this problem, since some of these variations, such as dopant fluctuations, are purely random. Although their relative effect decreases with the number of logic stages along a timing path, the current design approach, however, is to reduce the number of logic stages between registers, in order to increase the clock frequency. Also, traditional STA-based design optimization tends to create a large number of critical paths having the delay just slightly below the maximum allowable path delay. If statistical considerations are taken into account, the variation of the actual delay distribution increases with the number of critical paths [16]. Statistical design for digital circuits is a promising approach to handle larger process variations, especially OCVs. The goal is to treat these variations, which are random in nature, as statistical parameters during design, thus allowing a more accurate description, and eliminating the need for massive guard-banding. Moreover, sensitivities with respect to variations may be properly identified, allowing to performing statistical optimization. In the following Sections, some basic concepts of STA will be reviewed. Subsequently, Monte Carlo (MC) analysis, which represents a possible solution to process variations, will be discussed, along with the main algorithms and methodologies proposed in the literature for SSTA.

3.1. Static timing analysis

In digital circuits, it is required to compute an upper bound on the delay of all paths from the primary inputs to the primary outputs, irrespective of the input signals. Such upper bound is computed by means of a static simulation, known as static timing analysis (STA) [17]. STA is a highly efficient method to characterize the timing performance of digital circuits, to determine the critical path, and to obtain accurate delay information. Fig. 8 shows a simple circuit consisting of two banks of (ideal) flip-flops and four combinational blocks. In this example STA predicts the earliest time when FF2 can be clocked, while ensuring that valid signals are being latched into all flip-flops and registers. Before performing STA, each combinational block delay is pre-characterized. The delay from each input to each output pin is either described as an equation, or stored into a look-up table. Delay is a function of variables such as input slope, fanout, and output capacitive load. The pre-characterization phase consists of many circuit simulations at different temperatures, power supply voltages, and loading conditions. Delay data from these simulations are abstracted into a timing model for each block. The analysis is carried out in two phases. First, the delay of each signal is propagated forward through the combinational blocks, using the pre-characterized delay models and computing the wire delay, typically exploiting reduced-order macro-models, based on model order reduction (MOR) techniques of the original interconnects [18–20]. Thus, each signal is labeled with its latest arrival time where the correct digital value can be guaranteed. Next, the required arrival time is propagated backwards from the target bank of flip-flops (namely FF2 in the example). The required arrival time on a signal is the latest time the signal must have its correct value in order for the system to meet the timing requirements. The difference between the required arrival time and the actual arrival time for each signal is the signal slack. After the analysis, all signals are sorted according to their slack increasing order. If there is a negative slack on any of the signals, the circuit will not meet the performance requirements. The path with the minimum slack on all its signals is the critical path. The above analysis can be carried out with a minimum and maximum delay for each block. In this case, a set of early and late arrival times is computed for each signal. The early mode is computed using the best-case for the arrival times of all input signals to a block, while the late mode considers the most pessimistic scenario.

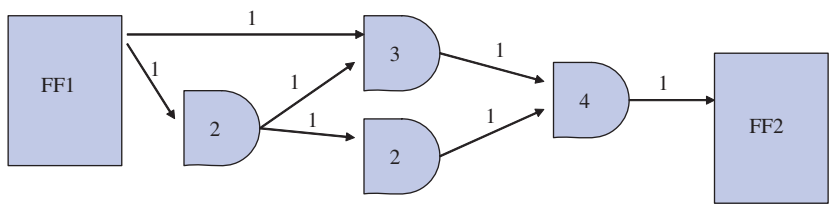

Fig. 8. Illustration of static timing analysis.

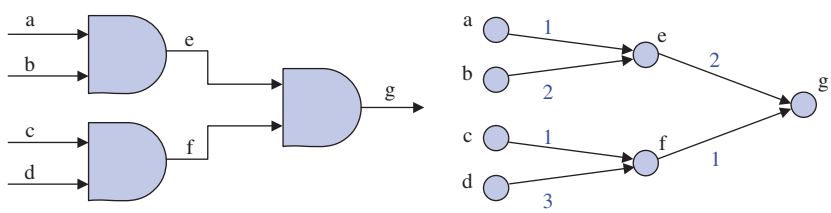

Fig. 9. A simple combinational circuit (left) and its corresponding timing graph (right).

3.1.1. Path-enumeration and block-oriented algorithms

In STA, the timing information contained in a combinational logic network is modeled with timing graph, which is a *Directed Acyclic Graph* (DAG), as shown in Fig. 9. A timing graph G corresponding to a logic network C consists of a set V of nodes and a set E of edges $G(V, E)$, such that every signal line in C is represented as a node in V and every input–output pair of every gate in C is represented as an edge in G. The signal propagation delay associated with an input–output pair is represented as a weight on the corresponding edge in G. Most methods adopted in STA for digital circuits can be divided into two major categories: path-enumeration (*path-based*), and block-oriented (*block-based*) techniques. Path enumeration is based on *depth-first* traversals of the timing graph. First, all topological paths are identified according to well-known algorithms, as illustrated in Fig. 10 (above). Then, the top K-critical paths are selected, and for each path the total delay is computed and compared against the required value. An efficient generation of the top K-critical paths is crucial to path-based approaches [21]. The path-based algorithms are well suited to handling correlations between gate delays and path sharing (i.e., *reconvergent fanouts*), but they have long run times, as the number of paths through a graph grows exponentially with the size of the graph. In contrast, block-based techniques do not generate paths, but work through a levelized timing graph in a *breadth-first* fashion. Basically, in the Program Evaluation and Review Technique (PERT) model [22], blocks are levelized and processed following their level order, as shown in Fig. 10 (below). Block-based algorithms are inherently linear in complexity, but their significant downside is the inability to handle correlations, such as between a clock path and a datapath.

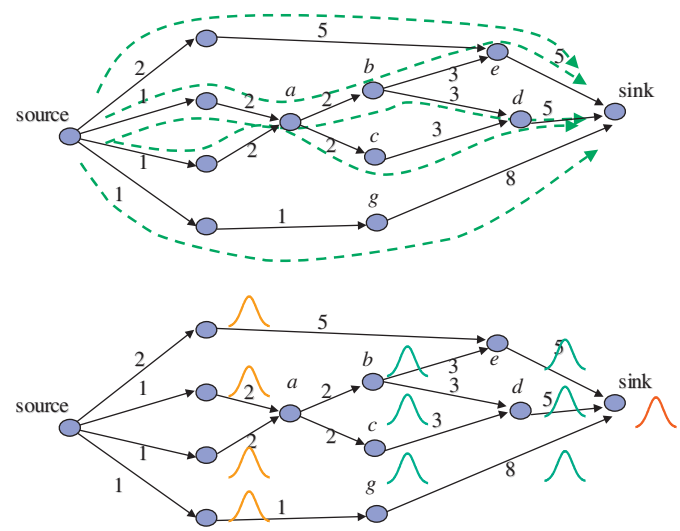

Fig. 10. Path-based (above) and block-based (below) timing graph traversal.

3.2. Monte Carlo methods

One approach for predicting the effects of parameter variations is MC analysis. It is a "brute force" method that never fails, and in some cases may be the only available option. It consists of several trials, each of which is a full-scale circuit simulation. On every simulation, each process parameter is sampled from its distribution, and then a STA is performed to obtain the output delay [23]. The procedure is repeated over thousands of trials, and the output delay distribution is derived from the collection of output delays. With a sufficient number of trials, the output distribution can be predicted with a measurable confidence. An estimation of the timing yield is then obtained by considering the fraction of samples for which the timing constraint is satisfied. MC-based techniques are inherently accurate as they do not involve any approximation. In the conventional approach, based on a fully random choice of the samples, the number of employed samples N is crucial. In fact, the runtime directly depends on N (leading to a loss of efficiency for large values of N), while the accuracy of the estimator for timing yield has large variance for small N (variance decreases proportionally to $\sqrt{N}$). In order to reduce the sample size for MC-based methods, several techniques were proposed in the literature, called *variance reduction techniques*. The exploitation of these methods for parametric yield estimation has been recently proposed in several works addressing the efficiency improvement of MC statistical timing analysis.

Techniques for efficient MC methods involve the estimation of the value of a definite finite-dimensional integral in the following form:

$$G = \int_{\Omega} g(X) f(X)\, \mathrm{d}X, \tag{1}$$

where Ω is a finite domain, X is a vector variable representing the process parameters, and $f(X)$ is the PDF on X. If $g(X)$ is a function that evaluates to 1 when the circuit delay is within the specifications and 0 otherwise, then the value of the integral G is the circuit yield. MC estimation for the value of G is obtained

by drawing a set of samples $X_1, X_2, \ldots, X_n$ from $f(X)$ and letting the estimator G_N be given by the following expression:

$$G_N = \frac{1}{N}\sum_{i=1}^{N} g(X_i). \tag{2}$$

The variance reduction techniques typically reduce the number of MC simulations required to accurately estimate (i.e., with small variance) the value of the finite integral (1) by means of expression (2). The work [24] focused on the *importance sampling* and the *control variates* techniques. The first method biases the choice of the samples from the process parameter space towards areas where the circuit delay violates the timing constraints (called important regions). Mathematically, the technique is based on drawing the samples for X from another distribution $\tilde{f}$ in order to reduce the variance of the estimator G_N. Integral (1) is then written as

$$G = \int_{\Omega}\left(\frac{g(X)f(X)}{\tilde{f}(X)}\right)\tilde{f}(X)\,\mathrm{d}X$$

and if $X_1, X_2, \ldots, X_n$ are drawn from $\tilde{f}$ instead of f, the new estimator is expressed as

$$\tilde{G}_N = \frac{1}{N}\sum_{i=1}^{N}\frac{g(X_i)f(X_i)}{\tilde{f}(X_i)}.$$

Ideally, the choice of $\tilde{f}$ that minimizes the variance of the estimator G_N is given by

$$\tilde{f}_{ideal}(X) = \frac{g(X)f(X)}{G},$$

but in practice $\tilde{f}_{ideal}$ cannot be realized since the value of G is not known a priori. Instead, a function $\tilde{f}$ "similar" to $\tilde{f}_{ideal}$ is typically used.

In the *control variates* approach, a function $h(X)$ that "correlates well" with $g(X)$ is used. The function h must be chosen so that the integral:

$$H = \int_{\Omega} h(X)f(X)\,\mathrm{d}X$$

can be evaluated with very low variance, e.g., it is known analytically and $\Delta(X) = g(x) - h(X)$ has a much smaller variance than $g(X)$ itself. Eq. (1) can be written as

$$\begin{aligned} G &= \int_{\Omega}(g(X) - h(X))f(X)\,\mathrm{d}X + \int_{\Omega} h(X)f(X)\,\mathrm{d}X \\ &= \int_{\Omega}\Delta(X)f(X)\,\mathrm{d}X + \int_{\Omega} h(X)f(X)\,\mathrm{d}X, \end{aligned}$$

while the estimator for G becomes

$$G_{cm} = H + \frac{1}{N}\sum_{i=1}^{N}\Delta(X_i).$$

Since H can be estimated with zero or very low variance, and all $\Delta(X_i)$ values (and therefore their contribution to the total variance) are very small, a variance reduction is then obtained.

In order to be effective, these techniques require a function that well approximates $g(X)$. In [24] the authors firstly defined the timing yield as an integral in the form of (1), by defining an indicator variable $I(S, X)$ that evaluates to 1 if the circuit delay does not meet the timing target, and 0 otherwise. The variable S represents the fixed design parameters for the circuit. Therefore, the yield computation can be expressed as in (1)

$$Loss(S) = 1 - Yield(S) = \int I(S,X)f(X)\,\mathrm{d}X.$$

Then, for estimating the timing yield, it was proposed to use the logical effort approximation to obtain a function that approximates $I(S, X)$ and has the mathematical properties required by the variance reduction methods. In [24] the control variates technique is used in conjunction with importance sampling; however, no experimental results were presented. The work in [25] presented an efficient formulation of the importance sampling method, called *mixture importance sampling*, for statistical SRAM design and analysis. To produce more samples in the important region, where the delay does not meet the target, the authors proposed to distort the (natural) sampling function by using an appropriate mixture of distributions, including a shifted Gaussian and a uniform distribution. The reported results demonstrated some efficiency and accuracy improvement against the standard MC analysis. A further application of the importance sampling technique to speed-up path-based MC simulations for statistical timing analysis was proposed in [26].

Another variance reduction technique suitable for parametric yield estimation is Latin Hypercube Sampling (LHS). The advantage of LHS over the importance sampling and control variates techniques is that is does not require any knowledge of the system under consideration, and is therefore general and scalable. LHS attempts to ensure that the chosen samples are spread more or less uniformly in the sample space. In a simple version, LHS generates N samples from a sample space of k random variables $X = [X_1, X_2, \ldots, X_k]$ in the following manner. The range of each variable is partitioned into N non-overlapping intervals of equal probability size $1/N$. One value is chosen at random from each of these N intervals for every variable, and the N values thus obtained for X_1 are randomly paired with the N values obtained for X_2. This results in N pairs that are combined randomly with the N values of X_3 to form N triplets. The procedure continues until N k-tuples are obtained. Fig. 11 illustrates LHS sampling algorithm for the three-variable case [27]. LHS achieves variance reduction

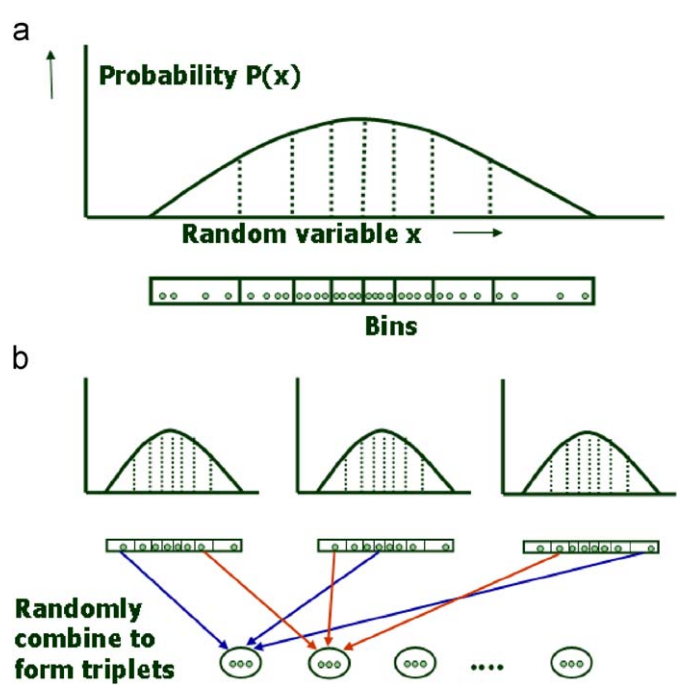

Fig. 11. Example of LHS sampling with $N = 8$, $k = 3$: (a) sampling of a variable in equal probability bins and (b) forming triplets by randomly combining individual samples [27].

in very general cases and can be effectively combined with other techniques for variance reduction. In [27], a Criticality Aware Latin Hypercube Sampling (CALHS) approach is introduced to improve the efficiency of MC-based statistical timing analysis. Timing criticality information is used to partition the process space into mutually exclusive strata. Then, the LHS technique determines an appropriate set of samples in these strata. By assuming that process variations can be represented as a linear combination of orthogonal random variables, and by assuming a linear relationship between the gate delay and the principal components of all the parameters and the uncorrelated random component (the validity of both the above assumptions will be discussed in the next section), the results in [27] showed about 7 × reduction in the number of samples compared to random sampling. Moreover, the MC-based SSTA with CALHS computed the 99th percentile circuit delay with about 50% less error than a traditional SSTA-based approach.

Another variance reduction technique is represented by the Quasi-Monte Carlo (QMC) method. The error bound to numerically estimate integral (1) by using a sequence of samples can be related to a mathematical measure of uniformity for the distribution of the points, called "discrepancy". This suggests that sequences with the smallest discrepancy should be used to evaluate the function in order to achieve the smallest possible error bound. Such sequences constructed to reduce discrepancy are called Low Discrepancy Sequences (LDS) and they are deterministic. QMC techniques are characterized by using LDSs to generate samples. However, their exploitation in SSTA is not straightforward, since when the problem dimension increases, there is degraded uniformity (*pattern dependency*, [28]). To minimize this effect, the concept of criticality of variables was introduced in [29], where a technique for variable ordering based on their criticality with respect to circuit delay is proposed. The variables are separated into critical, moderate, and non-critical ones. Then, the variance reduction techniques are applied where they are most effective. For the top-most critical variables, the stratified sampling technique is used, leading to faster accuracy. Only the top 2–5 variables are used to guide stratification since the number of strata increases exponentially with the number of variables. QMC methods are employed on the top-most to moderately critical variables for its fast convergence properties. Because of pattern dependency, only a limited number of variables are sampled with QMC. Therefore, on the non-critical variables, the LHS technique is adopted. This approach, called Stratification+Hybrid QMC (SH-QMC), achieved on average about 24 × reduction in the number of samples required for timing estimation compared to a random sampling approach. Moreover, SH-QMC is suitable for incremental timing analysis, when a fast recomputation of the circuit delay with small changes in the design is necessary. In fact, if the samples for SH-QMC on circuit C are reused for C' (C with small changes), then most samples need not be reevaluated to recompute the xth percentile delay; only those samples with a circuit arrival time close enough to the xth percentile delay of C need to be re-evaluated.

However, although these techniques improve the performance of MC-based SSTA, and some limitations can be discussed and possibly removed [30], there is a general agreement that more research is required to assess if MC methods can be effective for the timing yield estimation of large system-on-chip (SoC) designs.

3.3. Probabilistic analysis methods

While MC techniques are based on sample space enumeration, other methods explicitly model timing quantities such as delays, arrival times, and slacks as probability distributions; they are referred as *Probabilistic Analysis Methods*. The equivalent timing graph is probabilistic, and delays are random variables, as illustrated in Fig. 12. Therefore, the probability distribution of the circuit performance under the influence of parameter variations can be predicted with a single timing analysis. The problem of unnecessary risks, excessive number of timing analyses, and pessimism are all potentially avoided. Moreover, the WID variations, which are random in nature, are actually considered as statistical quantities during the analysis. Finally, other phenomena can be considered statistically such as [9]

- The inaccuracy of the model-to-hardware correlation can be treated statistically to reduce pessimism.
- Aging and fatigue effects such as negative bias temperature instability (NBTI), hot electron effects, and electromigration can be considered with probabilistic techniques.
- *Coupling noise* can be probabilistically integrated into a unified timing verification environment. However, coupling effects are typically not considered as variability sources. SSTA algorithms including coupling effects will be discussed in Section 4.7.

A typical SSTA tool accepts additional input information with respect to a traditional timing analyzer, including the sources of variation and their probability distributions, variances and co-variances. Moreover, it is possible to compute the dependence of the cell delay and slew on the sources of variability. The main output of the tool is the probability distribution of the slack and probabilistic diagnostics.

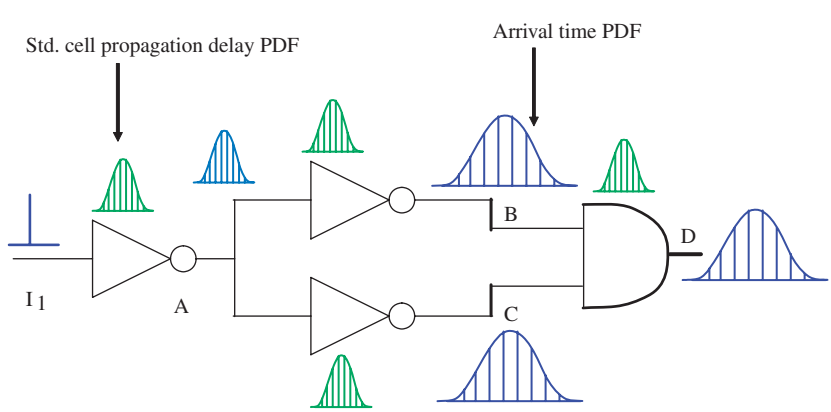

Fig. 12. A probabilistic timing graph.

3.4. Key challenges for statistical static timing analysis

Taking spatial correlations into account is a crucial requirement for SSTA [31]. There are several kinds of correlation that must be considered. The first ones are structural correlations introduced by different data paths sharing some standard cells, otherwise known as reconvergent fanouts. The second type of correlation is related to spatial proximity: devices and wires that are within the same layout region exhibit very similar parameter variations, because they are caused by the same manufacturing sources. For instance, standard cells close to each other are likely to have very small channel length variation; therefore, their delays are also quite similar. Moreover, it is very likely that transistors and interconnects within the same layout region also have similar temperature and power supply values. Hence, this type of correlation is known as spatial correlation.

Another challenge is represented by the delay modeling for cells and interconnects. While most process variations can be described by means of a normal distribution, this is not necessarily the case for the delay variations introduced by such process variations. In order to simplify calculations and reduce the overall computational effort for SSTA, most approaches assumed a linear dependency of delay on process variations. Recently, higher-order models have been proposed, while analytical modeling of gate-level behavior has not received much attention as yet. The propagation of delay distributions through a circuit represents another critical issue in SSTA. After the delay distribution of all circuit components has been modeled, the delay of an entire circuit needs to be computed. Operations of fundamental importance in block-based analysis are the sum and the max/min of random variables. In particular, for the max/min operation, it is computationally very expensive to determine the exact result. Therefore, most of the proposed approaches make the simplifying assumption that the result of these operations is also a normal distribution.

A critical topic is related to the different algorithmic approaches used to compute the delay distribution, i.e., path-based or block-based, which may differ significantly in terms of both accuracy and computational complexity. Due to the large computational effort necessary for path-based analysis, in [31] it was proposed first to run traditional STA, and then to analyze only the n-most critical paths accurately using SSTA. However, some potentially statistically critical paths may be missed, as illustrated in Fig. 13. This plot shows the probability that a given path is in the top 50 worst-case paths on a given die. The paths are ranked on the x-axis by margins (computed deterministically with worst-case STA) at the latching flip-flops. As shown in Fig. 13, several paths with rank higher than 100 show up in the top 50 paths for the block on 10% of the dies. This result demonstrates that deterministic timing analysis may not give an accurate path ordering [32]. All path-based methods have the fundamental limitation that the number of paths is too large and some heuristics must be used to limit the critical paths considered for detailed analysis. On the other hand, block-based approaches, while computationally more efficient, suffer from a lack of accuracy especially due to the statistical max/min operation. In the next section, the main approaches proposed in the literature addressing the challenges discussed above will be analyzed, focusing the attention on the block-based approach, which enables SSTA on multi-million gate designs in a reasonable amount of time.

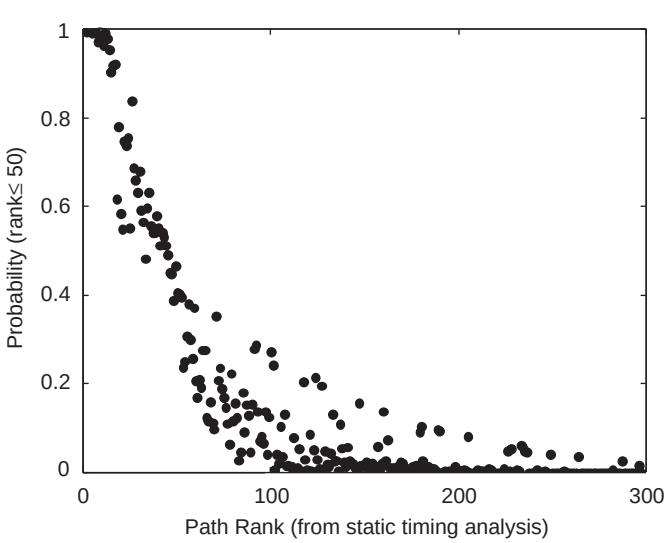

Fig. 13. Probability that a path is in the top 50 critical paths. Data from Monte Carlo analysis of a 90 nm microprocessor block [32].

4. Block-based statistical static timing analysis

One of the most useful approaches for circuit analysis and optimization is parameterized statistical timing analysis. This technique considers gate and wire delays as functions of the process parameters. Using this representation, parameterized statistical timing analysis computes circuit timing characteristics (arrival times, delays, timing slacks) as functions of the same parameters. Knowing explicit dependencies of timing characteristics on process parameters has two main advantages. First, by combining this information with the parameter statistics, we can compute the probability distribution of circuit delay and predict manufacturing yield. Then, this information can be used for circuit optimization, improving the design robustness and manufacturing line tailoring. In contrast, non-parameterized statistical timing analysis cannot compute relations between circuit timing characteristics and process parameters [33–36]. The most important works on parameterized SSTA using a block-based approach were proposed by Visweswariah et al. [37], and Chang and Sapatnekar [38]. The work of Visweswariah et al. was one of the first statistical timing methods that were exploited in an industrial tool by IBM, called EinsStat.

4.1. The canonical first-order delay model

Although there are several significant correlations in the timing variability of digital circuits, there are also some completely random sources of variation. For example, the dopant concentration density and oxide thickness variations from transistor to transistor in a nanometer technology can be considered as random. In order to account for both global correlations and independent randomness, the following canonical first-order delay model was proposed in [37] for all the timing quantities:

$$a_0 + \sum_{i=1}^{n} a_i \Delta X_i + a_{n+1} \Delta R_a. \tag{3}$$

It consists of a deterministic (mean or nominal value) portion a_0, a correlated (or global) portion: $\sum_{i=1}^{n} a_i \Delta X_i$, an independent (or local) portion: $a_{n+1} \Delta R_a$. In expression (3) the terms ΔX_i, $i = 1, \ldots, n$, represent the fluctuations of n global sources of variation X_i, centralized by subtracting their mean value: $\Delta X_i = X_i - \hat{X}_i$. Moreover, a_i, $i = 1, 2 \ldots, n$, are the sensitivities of gate delay (or other timing characteristics) to each of the global sources of variation,

ΔR_a is the variation of an independent random variable R_a from its nominal value, and a_{n+1} is the sensitivity of the gate delay (or other timing quantities) to uncorrelated variations. Since the sensitivity coefficients may be scaled, it can be assumed that X_i and R_a are normal Gaussian distributions $N(0, 1)$, with zero mean and unit variance. Therefore, the resulting delay (or other timing characteristics) is Gaussian, as it is expressed by a weighted sum (or linear combination) of Gaussian distributions. Obviously, since the model is obtained by considering the first-order terms of the Taylor expansion, it is valid only for small fluctuations of the process parameters. The above parameterized delay model allows the SSTA tool to determine the delay of a gate (wire) as a function not only of the traditional delay-model variables (like input slew and output load), but also as a function of the sources of variation. This canonical delay model is based on the sensitivities, which can be obtained by means of circuit simulations during a pre-characterization step. The parameterized delay model must be provided to the SSTA tool along with the sources of variation distributions, which are typically represented by a mean value and standard deviation. Any correlation between the sources of variation can be also specified.

4.2. Circuit delay calculation in block-based statistical timing analysis

In order to apply the block-based algorithm in statistical timing analysis, we must find the probability distributions of the *sum* (*difference*) and *max* (*min*) of a set of correlated Gaussian random variables, since the output delay of a multi-input gate shown in Fig. 14 can be calculated by

$$A_{out} = \max_{i=1}^{n}(A_i + D_i),$$

where n is the number of *fanins*. The *sum* of two random variables is a linear function; hence the *sum* of Gaussians is still a Gaussian distribution. In contrast, the *max* of two random variables is a nonlinear function, thus the *max* of two Gaussians in general is not Gaussian. Berkelaar [39] proposed a technique to approximate the result of *max* operation between Gaussians with a Gaussian distribution. The analytical expressions for both the mean and variance of the approximated *max* operation are reported in [40]. However, Berkelaar's approach is restricted to uncorrelated random variables, and to take correlations into account, a new approach was proposed by Tsukiyama et al. [41]. In this method, the *max* operation is approximated by a Gaussian, whose mean and variance can be computed analytically by using the Clark's results [42]. Given two random variables A and B and their Gaussian distributions $A = N(\mu_A, \sigma_A)$ and $B = N(\mu_B, \sigma_B)$ with a correlation coefficient $\rho = r(A, B)$, the mean and variance of $C = \max(A, B)$ are given by

$$\begin{aligned}\mu_C &= \mu_A\Phi(\beta) + \mu_B\Phi(-\beta) + \theta\varphi(\beta),\\ \sigma_C^2 &= (\mu_A^2 + \sigma_A^2)\Phi(\beta) + (\mu_B^2 + \sigma_B^2)\Phi(-\beta) + (\mu_A + \mu_B)\theta\varphi(\beta) - \mu_C^2,\end{aligned} \tag{4}$$

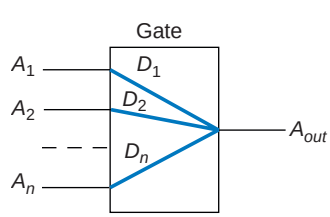

Fig. 14. General gate delay model.

where

$$\begin{aligned}&\theta = \sqrt{\sigma_A^2 + \sigma_B^2 - 2\rho\sigma_A\sigma_B}, \quad \beta = \left(\frac{\mu_A - \mu_B}{\theta}\right)\\ &\varphi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}, \qquad \Phi(y) = \int_{-\infty}^{y}\varphi(x)\,dx\end{aligned} \tag{5}$$

Clark's formulas (4) and (5) will not apply if $\sigma_A = \sigma_B$ and $\rho = 1$, but in this case, the *max* function is simply identical to the random variable with the largest mean value. Moreover, from [42], if γ is another normally distributed random variable with correlation coefficients $r(A, \gamma) = \rho_A$ and $r(B, \gamma) = \rho_B$, then the correlation between γ and C can be obtained by

$$r(C, \lambda) = \frac{\sigma_A\rho_A\Phi(\beta) + \sigma_B\rho_B\Phi(-\beta)}{\sigma_C}.$$

Therefore, the result of the *max* operation C is approximated to a Gaussian variable $C_N = N(\mu_C, \sigma_C)$. The first and the second moments of C are matched to obtain C_N, while the higher-order moments of C are ignored. This is the first and foremost source of inaccuracy in the approach. The nonlinearity of the *max* operation causes C to have an asymmetric density function, while the approximated Gaussian variable C_N has a symmetric density function. A quantification of the error introduced in the above approximation was derived in [43]. Given two random variables X and Y along with their PDFs, the error $\Xi_{X,Y}$ between the variables is defined as the total area under the non-overlapped region of their PDF. The work [43] proved that the approximation error in the *max* of any two Gaussians $A = N(\mu_A, \sigma_A)$ and $B = N(\mu_B, \sigma_B)$ can be estimated from the approximation error in the *max* of two derived Gaussians, one of which is the unit normal Gaussian and the other one is defined as $Z = N(\mu_Z, \sigma_Z) = N((\mu_A - \mu_B)/(\sigma_A), (\sigma_B/\sigma_A))$. The error $\Xi_{(C)(C_N)}$ is therefore a function of μ_Z, σ_Z and the correlation coefficient ρ. Since β (as defined in (5)) is a function of μ_Z, the error can be expressed as function of β, σ_Z and ρ. In [43] experiments were performed to study the dependency of the error $\Xi_{(C)(C_N)}$ on the above parameters. It was observed that $\Xi_{(C)(C_N)}$ decreases when one of the Gaussians dominates the other ($|\beta| \geqslant 3$), and increases when the Gaussians contribute almost equally to the max (β in the neighborhood of 0). $\Xi_{(C)(C_N)}$ is found to increase with decreasing σ_Z and is convex with respect to the correlation coefficient.

To increase the accuracy of the *max* computation, in [44] it was proposed an analytical approach that extends Clark's results to skew the normal distribution. Starting from a normal distribution with mean μ and variance σ given by

$$f(x) = \frac{1}{\sigma}\varphi\left(\frac{x - \mu}{\sigma}\right),$$

a skewed normal distribution can be computed from the normal distribution by scaling its left-half and right-half by factor γ and its inverse $1/\gamma$, respectively. Therefore, the skewed normal distribution can be written as follows:

$$f_\gamma(x) = \frac{2}{\sigma_l + \sigma_r}\left\{\varphi\left(\frac{x - \mu}{\sigma_l}\right)I_{(-\infty,\mu]}(x) + \varphi\left(\frac{x - \mu}{\sigma_r}\right)I_{(\mu,\infty)}(x)\right\}, \tag{6}$$

where $\sigma_l = (\sigma/\gamma)$, $\sigma_r = \sigma\gamma$, and $I_A(x)$ is the Indicator function: $I_A(x) = 1$ if $x \in A$, 0 otherwise. If the skewness parameter γ is greater (less) than unity, then $f_\gamma(x)$ is positively (negatively) skewed, while for $\gamma = 1$, (6) reduces to the normal distribution. Function (6) is both continuous and differentiable, and it is completely defined by only three parameters: μ, σ, and γ. Given a generic arrival time distribution characterized by its mean μ_γ, variance σ_γ, and skewness Sk_γ, it can be easily mapped to a skewed normal distribution by moment matching. As derived in [44], the skewness of distribution defined by the ratio of the third centered moment and cubed deviation is only function of the

parameter γ. Therefore, for a given Sk_γ, γ can be efficiently computed either by using pre-computed look-up tables or by using numerical techniques. Then, using γ, σ_γ, and μ_γ, the following two equations matching the first two moments can be solved for parameters σ and μ, respectively:

$$\mu_\gamma = \mu + \sqrt{2/\pi}(\gamma - 1/\gamma)\sigma,$$
$$\sigma_\gamma^2 = \frac{(\pi\gamma^4 - 2\gamma^4 - \pi\gamma^2 + 4\gamma^2 + \pi - 2)\sigma^2}{\pi\gamma^2}.$$

In order to analytically express the *max* function of two correlated arrival time random variables X and Y, their joint probability density function (JPDF) must be known. In [42], the following bivariate normal distribution for two operands X and Y was used:

$$f(x,y) = \frac{1}{2\pi\sigma_x\sigma_y}\varphi\left(\frac{x-\mu_x}{\sigma_x}, \frac{y-\mu_y}{\sigma_y}\right),$$
$$\varphi(x,y) = \frac{1}{\sqrt{1-\rho^2}}\mathrm{e}^{-(x^2-2\rho xy+y^2)/(2(1-\rho^2))}.$$

Therefore, similarly to the univariate skewed normal, in [44] the authors added two inverse scale parameters γ_x and γ_y to introduce skewness in the bivariate distribution. Then, for this bivariate skewed normal distribution, they derived analytical results for efficiently computing the approximate moments of the *max* of X and Y based on the original derivation given in [42]. From these moments the mean, variance, and skewness of the maximum can be computed. Therefore the proposed approach can be exploited in existing SSTA tools based on Clark's result, taking into account skewness the of X and Y in addition to mean and variance of the arrival time distribution.

The canonical first-order delay model by Visweswariah et al. [37] described in Section 4.1, uses the Clark's formulas (4) and (5) along with the concept of tightness probability to determine the distribution of the max of two arrival times. Given two random variables X and Y, the tightness probability T_X of X is the probability that X is larger than (or dominates) Y. Given n random variables, the tightness probability of each variable is the probability that it is larger than all the others. If T_X is the tightness probability of X, then the tightness probability of Y is: $T_Y = 1 - T_X$. Given two timing quantities A and B expressed in canonical first-order form (3):

$$A = a_0 + \sum_{i=1}^{n} a_i\Delta X_i + a_{n+1}\Delta R_a$$
$$B = b_0 + \sum_{i=1}^{n} b_i\Delta X_i + b_{n+1}\Delta R_b$$

it can be shown that the variances σ_A, σ_B, and the correlation coefficient ρ can be computed in linear time as

$$\sigma_A = \sqrt{\sum_{i=1}^{n+1} a_i^2},$$
$$\sigma_B = \sqrt{\sum_{i=1}^{n+1} b_i^2},$$
$$\rho = \frac{\sum_{i=1}^{n} a_i b_i}{\sigma_A\sigma_B},$$
$$\mathrm{cov}(A,B) = \sum_{i=1}^{n} a_i b_i.$$

Moreover, in [37] by using Clark's formulas (4) and (5), the probability that A is larger than B, i.e., the tightness probability T_A, and the mean and variance of max(A, B) can also be expressed analytically as

$$T_A = \Phi\left(\frac{a_0 - b_0}{\theta}\right),$$
$$E[\max(A,B)] = a_0 T_A + b_0(1 - T_A) + \theta\varphi\left(\frac{a_0 - b_0}{\theta}\right),$$
$$\mathrm{var}[\max(A,B)] = (\sigma_A^2 + a_0^2)T_A + (\sigma_B^2 + b_0^2)(1 - T_A) + (a_0 + b_0)\theta\varphi\left(\frac{a_0 - b_0}{\theta}\right) - \{E[\max(A,B)]\}. \quad (7)$$

Therefore, the tightness probability, expected value, and variance of the *max* operation can be computed analytically and efficiently. The CPU time for this operation increases only linearly with the number of sources of variation. In order to further propagate through the timing graph the result of the *max* operation, we need to express $C = \max(A, B)$ back into canonical form. However, since the *max* of random variables is a nonlinear function, $C = \max(A, B)$ cannot be expressed exactly in canonical form. The key idea in Visweswariah's approach is to use the tightness probability concept to compute the statistical approximation C_{appr} of $C = \max(A, B)$. Tightness probability of timing quantity A (considered as a random variable), and expected value and variance of max(A, B) are given in (7). Tightness probabilities can be interpreted in the space of the sources of variation. If one random variable has a 0.3 tightness probability, then in 30% of the weighted volume of process space it is larger than the other variable, and in the other 70% the other variable is larger. The weighting factor is the JPDF of the underlying sources of variation. In traditional STA, C would take the largest value between A and B, and the characteristics of the dominant edge determining the arrival time C are preserved. This is similar to having a tightness probability of 100% and 0%. In the probabilistic domain, the characteristics of $C = \max(A, B)$ are determined from A and B *in the proportion of their tightness probabilities*. Therefore, we can express the canonical form of the approximation C_{appr} of the $C = \max(A, B)$ operation as $C_{appr} = c_0 + \sum_{i=1}^{n} c_i\Delta X_i + c_{n+1}\Delta R_c$, and the sensitivities c_i are given by

$$c_i = T_A a_i + (1 - T_A)b_i, \quad i = 1, 2, \ldots, n, \quad (8)$$

where a_i and b_i are the sensitivities of A and B, respectively, and T_A is the tightness probability of A. The mean of the distribution of $C = \max(A, B)$ is preserved when converting it into canonical form C_{appr}. The only remaining quantity to be computed is the independent random part of the canonical form and its sensitivity c_{n+1}. This is done by matching the variance of the canonical form to the variance computed analytically with (7), i.e., making the variance of C_{appr} equal to the variance of $C = \max(A, B)$. Thus, the first two moments of the real distribution are always matched in the canonical form. Moreover, the coefficients preserve the correct correlation to the global sources of variation as suggested in [9] and are similar to the coefficients computed in [38]. The covariance between $C = \max(A, B)$ and any random variable Y can be expressed in terms of the covariance between A and Y and B and Y as

$$\mathrm{cov}(C,Y) = \mathrm{cov}(A,Y)T_A + \mathrm{cov}(B,Y)(1 - T_A).$$

If we consider the random variable Y as one of the global sources of variation ΔX_i, $i = 1, \ldots, n$, and by observing that cov(A, ΔX_i) = a_i and cov(B, ΔX_i) = b_i, we obtain

$$\mathrm{cov}(C,\Delta X_i) = a_i T_A + b_i(1 - T_A)$$

and by assuming that C is normally distributed we obtain the sensitivities c_i (8). However, the covariance of the independent sources of variations ΔR_a and ΔR_b is not preserved.

The computation of a *two*-variable *max* function can be extended to *n*-variable *max* by repeating the computation of the *two*-variable case recursively, as proposed by Chang and

Sapatnekar [38]. The method is outlined in Fig. 15. However, the correlation (i.e., covariance) between the independent sources of variations (ΔR_a in canonical first-order form (3)) is not preserved. Moreover, during the recursive computation of n-variable *max* function, some inaccuracy can be introduced since the *max* is approximated by a normal distribution even though it is not normal. Such inaccuracy is exacerbated when proceeding with further recursive calculations. Therefore, as the number of variable increases, a larger error can be introduced. Moreover, the loss in accuracy of the final result is dependent on the ordering of the pair-wise *max* operations. The *max* operation on n Gaussians is analogous to the construction of a binary tree with n leaves such that each internal node computes the *max* of its two children. In [43] the above tree is referred as *Max Binary Tree* (MBT). Novel approaches for constructing good MBTs to reduce the *max* of n Gaussian inaccuracy have been proposed and analyzed in [43]. The experimental results of the proposed methods showed an accuracy improvement in variance estimation up to 50% against to the traditional approach.

The *sum* operation between two random variables (timing quantities) in canonical form, $D = A+B$, can be easily expressed in canonical form

$$D = (a_0 + b_0) + \sum_{i=1}^{n}(a_i + b_i)\Delta X_i + \sqrt{a_{n+1}^2 + b_{n+1}^2}\Delta R_d. \tag{9}$$

Therefore, by replacing the *sum* (*difference*) and *max* (*min*) operations with probabilistic equivalents, and by re-expressing the result in canonical form after each operation, SSTA can be carried out by a standard forward and backward propagation through the timing graph.

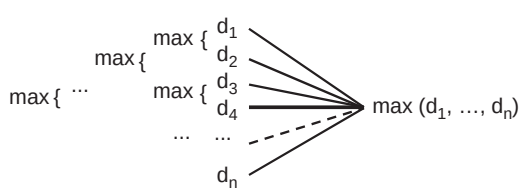

Fig. 15. Recursive computation of n-variable *max* function.

It is important to notice that the canonical first-order delay model (3) employed for all timing quantities allows to considering both global correlations and independent randomness, but it does not take into account the spatial correlations, which can be handled by means of derating factors. However, considering the spatial correlations by means of derating factors will yield inaccurate results in statistical timing analysis, which might be either pessimistic or risky. As such, spatial correlations must be included, and different modeling techniques will be discussed in the next section.

4.3. Spatial correlation modeling

Not every timing quantity depends on all global sources of variation, and the works [38,45,46] suggest methods for modeling parameter variations by having the delay of gates and wires in physically different die regions depending on different sets of random variables. The approach proposed in [45] is mainly focused on device channel length variability, but it can be straightforwardly extended to other process variations. The total channel length $L_{total,k}$ of device k is the algebraic sum of *nominal* channel length, *inter-die* channel length variation, and *intra-die* channel length variation:

$$L_{total,k} = L_{nom} + \Delta L_{inter} + \Delta L_{intra,k}, \tag{10}$$

where ΔL_{inter} and $\Delta L_{intra,k}$ are random variables, and L_{nom} represents the mean of the channel length across all possible dies which is equal to the nominal value of the device channel length. All devices on a die share one variable ΔL_{inter} for the inter-die component of their total channel length variation, which represents a variation of the *mean* of all the devices of a particular die. $\Delta L_{intra,k}$ is the variation of an individual device from this die *mean*. If the spatial correlation of intra-die variations is not considered, then each device is represented with a separate independent random variable $\Delta L_{intra,k}$, where all random variables $\Delta L_{intra,k}$ have identical probability distributions. Based on the assumption that for small variations the change in gate delay is linear with respect to the change in channel length, the delay of the k-gate can be expressed as

$$d_k = D_{nom} + \alpha\Delta L_{inter} + \alpha\Delta L_{intra,k}, \tag{11}$$

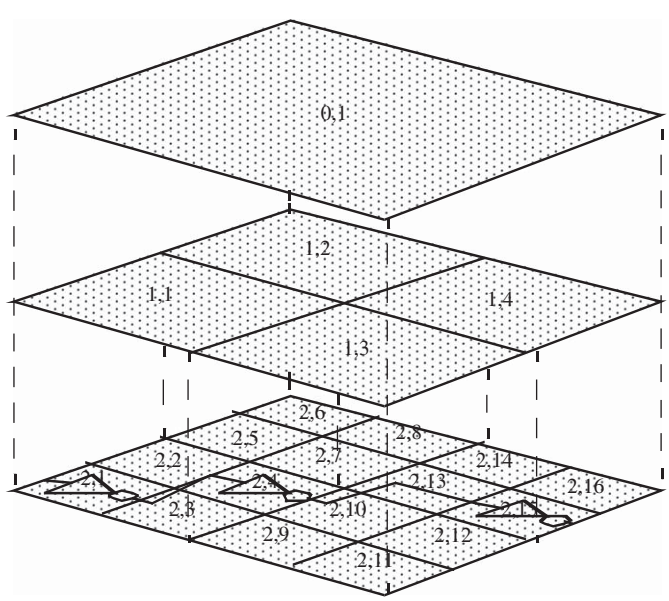

Fig. 16. Spatial correlation modeling with quad-tree partitioning [45].

where α is the sensitivity of the delay with respect to the channel length computed at the nominal device channel length. In (10) the intra-die variation of channel length is modeled by assigning an independent random variable for each gate. However, in presence of spatial correlations, these random variables become dependent, thus greatly complicating the analysis. Therefore, the following approach was proposed in [45]. The die area is divided into regions using a multi-level quad-tree partitioning, as shown in Fig. 16. For each level l, the die area is partitioned into 2^l-by-2^l squares, where the first or top level 0 has a single region for the entire die and the last or bottom level m has 4^m regions. Subsequently, an independent random variable $\Delta L_{l,r}$ is associated to each region (l, r) to represent a component of the total intra-die device channel length variation. The variation of gate k is then composed as the sum of intra-die components $\Delta L_{l,r}$, where level l ranges from 0 to m and the region r at any particular level is the region the intersects with the position of gate k. Hence, for the gate in region (2,1) in Fig. 16, the components of intra-die device length variation are $\Delta L_{0,1}$, $\Delta L_{1,1}$, $\Delta L_{2,1}$. The intra-die device channel length of gate k is thus defined as the sum of all random variables $\Delta L_{l,r}$ associated with a gate:

$$\Delta L_{intra,k} = \sum_{0 \leqslant l \leqslant m, r \text{ intersects } k} \Delta L_{l,r} + \Delta L_{random,k}, \tag{12}$$

where the last term in (12) is an independent random variable, assigned to each gate to model uncorrelated delay variation. The sum of all random variables $\Delta L_{l,r}$ associated with a gate always adds up to the total intra-die channel length variation. Hence, all random variables associated with a particular level are assigned the same probability distribution, and the total WID variability is divided among the different levels. Using this model, gates within close proximity of each other have many common intra-die channel length components resulting in a strong intra-die channel length correlation. In contrast, gates far apart on a die share few common components, and therefore have a weaker correlation. For the three gates in regions (2,1), (2,4) and (2,15) in Fig. 16, the intra-die channel length variation is expressed as

$$\begin{aligned} \Delta L_{intra,1} &= \Delta L_{2,1} + \Delta L_{1,1} + \Delta L_{0,1} + \Delta L_{random,1}, \\ \Delta L_{intra,2} &= \Delta L_{2,4} + \Delta L_{1,1} + \Delta L_{0,1} + \Delta L_{random,2}, \\ \Delta L_{intra,3} &= \Delta L_{2,15} + \Delta L_{1,4} + \Delta L_{0,1} + \Delta L_{random,3}. \end{aligned} \tag{13}$$

We can observe from (13) that gates in squares (2,1) and (2,4) are strongly correlated, as they share the common variables $\Delta L_{1,1}$ and $\Delta L_{0,1}$. On the other hand, gates in squares (2,1) and (2,15) are weakly correlated as they share only the common variable $\Delta L_{0,1}$. It is worth noticing that $\Delta L_{0,1}$ associated with the region at the top level of the hierarchy is equivalent to the inter-die device length ΔL_{inter} since it is shared by all gates on the die. We can control how quickly the spatial correlation diminishes as the separation between two gates increases by correctly allocating the total intra-die device length variation among the different levels. If the total intra-die variance is largely allocated to the bottom levels, and the regions at top levels have only a small variance, there is less sharing of device channel length variation between gates that are far apart and the spatial correlation will decrease quickly. The results will yield results that are close to uncorrelated intra-die analysis. On the other hand, if the total intra-die variance is predominantly allocated to the regions at the top levels of the hierarchy, then even gates that are widely spaced apart will still have a significant correlation. This will yield results that are close to the traditional approach where all gates are perfectly correlated and the intra-die device length variation is zero.

Based on the above model for intra-die spatial correlation, (11) and (12) can be combined obtaining the following expression of the gate delay:

$$d_k = D_{nom} + \alpha \left(\Delta L_{inter} + \sum_{0 \leqslant l \leqslant m, r \text{ intersects } k} \Delta L_{l,r} + \Delta L_{random,k} \right). \tag{14}$$

It is important to observe that all random variables in (14) are *independent* random variables, which greatly simplify the analysis. Finally, to further simplify expression (14), it can be re-written using a more general form as follows:

$$d_k = D_{nom} + \sum_i \alpha_i L_i + \Delta D_{random,k}, \tag{15}$$

where L_i and $\Delta D_{random,k}$ are random variables and α_i are constants. $\Delta D_{random,k}$ is the random delay due to uncorrelated intra-die channel length variation. The variables L_i correspond to one of the random variables in the proposed model, such as ΔL_{inter} and $\Delta L_{l,r}$. The sum is taken over all random variables present in the model and $\alpha_i = \alpha$ for the random variable ΔL_{inter} and for the random variables $\Delta L_{l,r}$ associated with the gate, based on its position on the die. For all other i, $\alpha_i = 0$. By using (15) the delay of a gate can be expressed as a sum of independent random variables. The model can be extended to the other sources of variation, re-obtaining the canonical first-order delay model.

To model the intra-die spatial correlations of process parameters, in [38] the die region is partitioned into $n_{row} \times n_{col} = n$ grids, as shown in Fig. 17. Since devices (wires) close to each other are more likely to have similar characteristics than those placed far away, this approach assumes perfect correlation among the devices (wires) in the same grid, high correlations among those in close grids, and low or zero correlation in far-away grids. For example, in Fig. 17 gates a and b are located in the same grid square, and it is assumed that their parameter variations (such as the variation of their gate length) are always identical. Gates a and c lie in neighboring grids, and their parameter variations are not identical but highly correlated due to their spatial proximity (for example, when gate a has a larger than nominal channel length, it is highly probable that gate c will have a larger than nominal channel length, and less probable that it will have a smaller than nominal channel length). On the other hand, gates a and d are far away from each other, and their parameters may be uncorrelated (i.e., when gate a has a larger than nominal channel length, the channel length for d may be either larger or smaller than nominal). Under this model, the parametric variation for a spatially correlated parameter in a single grid at location (x, y) can be modeled using a single random variable $p(x, y)$. In total, this representation requires n random variables for each parameter, where each random variable represents the value of the parameter in one of the n grids, and a covariance matrix of size $n \times n$ representing the spatial correlations among the grids.

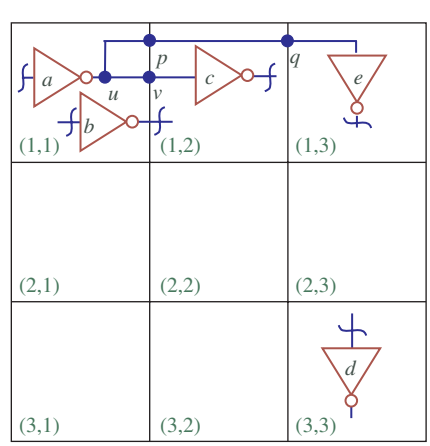

Fig. 17. Grid model for spatial correlation.

The covariance matrix can be determined from data extracted from manufactured wafers [47]. However, if real silicon data is not available, the correlation matrix can also be derived from the spatial correlation model proposed in [45,46].

It is believed that the correlation model proposed in [38] is more general than the model described in [45,46], since it is purely based on neighborhood. For example, consider the case in Fig. 18, where the 4×4 grids are numbered according to the quad-tree partitioning of Fig. 16. Following the model proposed in [38], the intra-die device length in grid (2,8) has equal correlations with that in grid (2,6) and (2,14), while by the model described in [45] it will have higher correlation with grid (2,6) than grid (2,14), i.e., the correlations are uneven at the two neighbors of grid (2,8), as summarized in

$$\begin{aligned} \Delta L_{\text{intra},a} &= \Delta L_{2,6} + \Delta L_{1,2} + \Delta L_{0,1} + \Delta L_{random,a}, \\ \Delta L_{\text{intra},c} &= \Delta L_{2,8} + \Delta L_{1,2} + \Delta L_{0,1} + \Delta L_{random,c}, \\ \Delta L_{\text{intra},e} &= \Delta L_{2,14} + \Delta L_{1,4} + \Delta L_{0,1} + \Delta L_{random,e}. \end{aligned} \tag{16}$$

We can observe from (16) that gates in squares (2,6) and (2,8) are strongly correlated, as they share the common variables $\Delta L_{1,2}$ and $\Delta L_{0,1}$. On the other hand, gates in squares (2,8) and (2,14) are weakly correlated as they share only the common variable $\Delta L_{0,1}$.

Another approach for spatial correlation modeling was proposed in [48]. A uniform grid is imposed on the placed netlist to partition the gates into spatial regions, as shown in Fig. 19, similarly to the technique proposed in [38]. The variation of a process parameter P can be represented as a linear combination of four independent random components P_1, P_2, P_3, and P_4, with zero mean and finite variance, which are random variables corresponding to the four corners of the chip (as depicted in Fig. 19). For any gate j, the corresponding parameter P_j can be modeled as

$$P_j = a_0 + a_1 P_1 + a_2 P_2 + a_3 P_3 + a_4 P_4, \tag{17}$$

where a_0 is the nominal value of parameter P_j. For any placed gate j we can compute the grid-based radial distance from the four corners of the placement, i.e., R_1, R_2, R_3, and R_4 in Fig. 19. The coefficients a_1, a_2, a_3, and a_4, in (17) can be computed by using these radial distances with an appropriate function $H(R)$ as follows:

$$a_1 = H(R_1); \quad a_2 = H(R_2); \quad a_3 = H(R_3); \quad a_4 = H(R_4). \tag{18}$$

The random variables P_1, P_2, P_3, and P_4 can have any arbitrary distributions, depending on the distribution of the parameter P_j. Hence, if two gates are far apart, they will have different contributions from the four components P_1, P_2, P_3, and P_4, and will have a weak correlation. In contrast, if they are placed close by, the four coefficients (18) will be similar, and a stronger spatial correlation will exist between them. This approach to model the spatial correlations is similar to the method proposed in [46]. However, in [46] the number of underlying variables to capture the spatial correlations is potentially higher, where in the approach proposed in [47] only four variables are necessary for each parameter. The importance of including spatial correlations in statistical timing analysis was demonstrated in [46], where ignoring such correlations may yield an under estimation of the computed variability.

The correlation models proposed in [38,45] were analyzed in [49], based on the critical dimension (CD) data obtained through electrical linewidth measurements (ELM) of a 130 nm test chip, consisting of 8 different test structures (various densities and orientations of polysilicon lines with OPC included), where 5 different wafers were investigated, each wafer containing 23 fields, and each field including 308 measurement points: 14 points in the horizontal direction and 22 points in the vertical direction. It was demonstrated that correlation is not monotonically decreasing with distance, as shown in Fig. 20, where it is evident that correlation vs. horizontal distance is different from correlation vs. the vertical distance (distance is not the key component to correlation, which is typically stronger along a particular axis). Moreover, it was reported that the number of

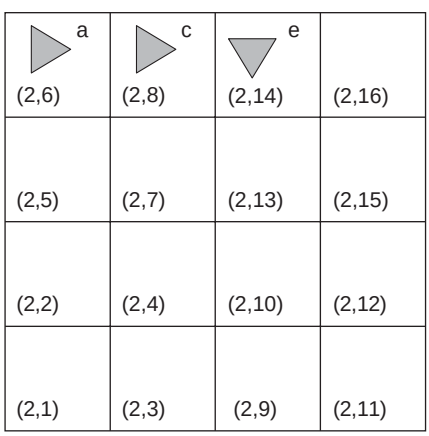

Fig. 18. Quad-tree partitioning (level 2).

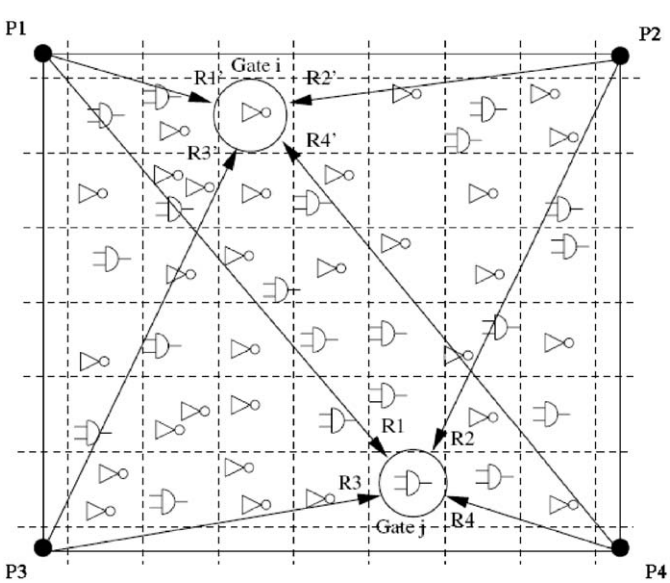

Fig. 19. Grid-based radial spatial correlation model [48].

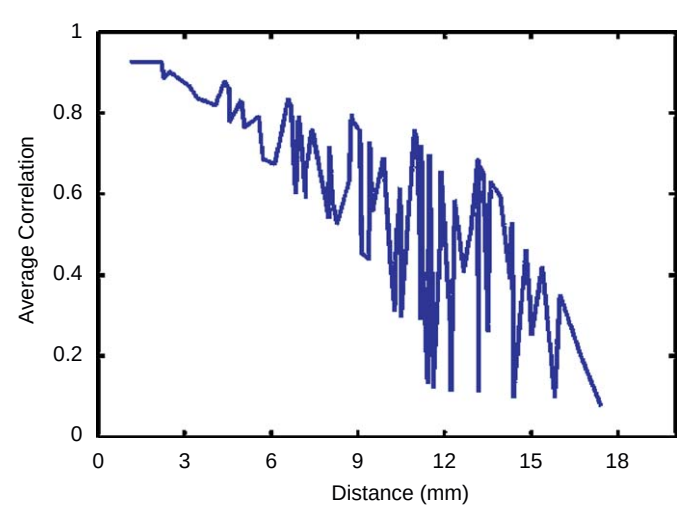

Fig. 20. Average correlation vs. distance [49].

principal components (from Principal Component Analysis) necessary to obtain accurate results with the grid-based approach presented in [38] is about 3, while for the quad-tree method [45] any number of levels above 3 did not give any significant improvement in terms of accuracy. The results presented in [49] demonstrate that both the grid-based approach [38] and the quad-tree method [45] provide an accurate estimation of the actual mean and variance of the circuit delay distributions. However, another interesting result reported in [49] is that also much simpler models (i.e., the die-to-die plus random model) for spatial correlations can yield a good accuracy, within a few percent of the grid-based models.

4.4. Orthogonal transformations of correlated random variables

In SSTA, when both the spatial correlations and the structural correlations due to *reconvergent fanouts* are taken into account, the overall correlation composition becomes very complicated. To make this problem tractable, in [38] the principal component analysis (PCA) technique is used to transform a set of correlated parameters into an uncorrelated set. Given a set of correlated random variables $\vec{X}$ with a covariance matrix R, PCA can transform the set $\vec{X}$ into a set of mutually orthogonal random variables $\vec{X}'$, such that each member of $\vec{X}'$ has zero mean and unit variance. The elements of the set $\vec{X}'$ are called *principal components* (PCs) in PCA, and are mathematical abstractions that cannot be directly measured. The size of $\vec{X}'$ is no larger than the size of $\vec{X}$, and any variable $x_i \in \vec{X}$ can be expressed in terms of the PCs $\vec{X}'$ as

$$x_i = \left(\sum_j \sqrt{\lambda_j} v_{ij} x'_j \right) \sigma_i + \mu_i,$$

where $x'_j \in \mathbf{X}'$ is a PC, λ_j is the jth eigenvalue of the covariance matrix R, v_{ij} is the ith element of the jth eigenvector of R, and σ_i and μ_i are the mean and standard deviation of x_i, respectively. For instance, let $\vec{L}_g$ be a vector of random variables representing transistor channel length fluctuations in all grids of Fig. 17, and the set of random variables is of multivariate normal distribution with covariance matrix R_{L_g}. Let $\vec{L}'_g$ be the set of PCs computed with PCA. Then any random variable $L^i_g \in \vec{L}_g$ representing the variation of transistor channel length in the ith grid can be expressed as linear function of the PCs:

$$L^i_g = \mu_{L^i_g} + a_{i1} l'^1_g + \cdots + a_{it} l'^t_g,$$

where $\mu_{L^i_g}$ is the mean of L^i_g, l'^i_g is a PC in $\vec{L}'_g$, all l'^i_g are independent with zero mean and unit variance, and t is the total number of PCs in $\vec{L}'_g$. In this way, any FEOL and BEOL process random variable can be expressed as a linear function of the corresponding principal components.

Hence, by assuming that different types of process parameters are uncorrelated and by approximating the delay linearly using a first-order Taylor expansion, gate and interconnect delays are random variables that can be expressed as a linear combination of PCs of all relevant FEOL and BEOL process parameters:

$$d = d_0 + \sum_{i=1}^{m} k_i p'_i, \tag{19}$$

where $p'_i \in \vec{P}'$, $\vec{P}'$ is the union of the sets of principal components of each relevant process parameters, m is the size of $\vec{P}'$ and all the PCs p'_i in (19) are independent. Since all p'_i are orthogonal random variables with zero mean and unit variance, the variance of d in (19) can be simply computed as

$$\sigma^2_d = \sum_{i=1}^{m} k^2_i, \tag{20}$$

while the covariance between d and any PC p'_i is given by

$$\mathrm{cov}(d, p'_i) = k_i \sigma^2_{p'_i} = k_i. \tag{21}$$

Moreover, if d_i and d_j are two random variables expressed in terms of PCs as

$$d_i = d^0_i + \sum_{r=1}^{m} k_{ir} p'_r,$$

$$d_j = d^0_j + \sum_{r=1}^{m} k_{jr} p'_r$$

their covariance can be computed by

$$\mathrm{cov}(d_i, d_j) = \sum_{r=1}^{m} k_{ir} k_{jr}.$$

In the work presented in [38], the above properties of delay in the form of Eq. (19) are used to find the distribution of circuit delay. The approach described in [38] to compute the *max* function of n normally distributed random variables is an extension of the method proposed in [40], which only considered uncorrelated random variables. In [38] the Gaussian distribution is used to approximate the *max* function $d_{max} \sim N(\mu_{max}, \sigma_{max})$ by means of a linear combination of all PCs as

$$d_{max} = \mu_{max} + \sum_{j=1}^{m} a_j p'_j. \tag{22}$$

Therefore, determining the approximation for d_{max} is equivalent to finding μ_{max} and all the coefficients a_j. From (21) the coefficient a_j equals to $\mathrm{cov}(d_{max}, p'_j)$ and the variance of d_{max} (22) can be expressed by means of (20) as

$$s^2_0 = \sum_{j=1}^{m} a^2_j = \sum_{j=1}^{m} \mathrm{cov}^2(d_{max}, p'_j). \tag{23}$$

Since (23) is an approximation, to reduce the difference between s^2_0 and the actual variance σ^2_{max} of d_{max}, the value a_j can be normalized as

$$a_j = \mathrm{cov}(d_{max}, p'_j) \frac{\sigma_{max}}{s_0}.$$

Hence, to find the linear approximation for d_{max} the values of μ_{max} and σ_{max} and $\mathrm{cov}(d_{max}, p'_j)$ are necessary. Those values can be obtained by using the Clark's formulas (4) and (5). This approach has similarities with [37], as they are both based on Clark's result; they differ in the fact that [37] uses its sensitivity to match variance while [38] scales all sensitivities to match variance (and thus it loses some correlation information).

Finally, in [38] an extension to consider also the intra-die spatially uncorrelated parameters was proposed. To model the intra-die variation of spatially uncorrelated parameters a separate random variable is used for each gate (wire), instead of a single random variable for all gates (wires) in the same grid for spatial correlated parameters. After each *sum* or *max* operation the random variations for spatially uncorrelated parameters are merged into one random variable. Hence, only one independent random variable is kept for all intra-die variations of spatially uncorrelated parameters. This technique of adding an independent random variable to the standard form of timing quantities is similar to [37]. However, in the approach presented in [38], the structural correlations due to spatially uncorrelated parameters cannot be handled.

4.5. Canonical form generalization

As it was discussed in the previous sections, one of the most promising approaches for circuit analysis and optimization taking

into account parameter variability is parameterized SSTA. This technique considers gate and wire delay D as function of process parameters X_i:

$$D = D(X_1, X_2, \ldots, X_n), \tag{24}$$

and Fig. 21 shows a graphical illustration of expression (24) for two process parameters. Using this description, parameterized SSTA computes circuit timing characteristics A (arrival and required arrival times, delays, timing slacks) as a function of the same process parameters:

$$A = A(X_1, X_2, \ldots, X_n), \tag{25}$$

Parameterized SSTA [37,38] assumes that *all parameters have independent normal Gaussian probability distributions and affect gate delays linearly*. The independence can be achieved by PCA. According to this assumption, gate delays are represented in first-order canonical form (3), where Fig. 22 shows the canonical form for one process parameter. In the case of multiple process parameters, the canonical form is represented by a hyper-plane defining the timing quantity (25) as a linear function of process parameters and two parallel hyper-planes bounding the 3σ region of uncertainty for the uncorrelated variation.

The assumption about the linear Gaussian nature of process parameters is very convenient for SSTA, since it allows the use of analytical formulas for computing canonical forms, thus making statistical timing analysis practical. Unfortunately, some process parameters have significantly non-Gaussian probability distributions. For example, via resistance is known to have an asymmetric probability distribution, and the dopant concentration density is also observed to be well-modeled by a Poisson distribution. Hence, a normality assumption may lead to errors. Moreover, the linear approximation is justified by small variations, but with critical feature size shrinking, the process variations are becoming larger and linear approximation is not accurate enough. For instance, delay dependence on transistor channel length (L_{eff}) is essentially nonlinear, and assuming linear dependency can result in substantially inaccurate results [50]. Furthermore, there is a nonlinearity source coming from the *max* operation, which generates non-Gaussian delay distribution even if the input operands are Gaussian distributions. The obvious way to handle process parameters that have non-Gaussian distributions and/or affect gate delay nonlinearly is to apply efficient numerical-integration techniques [31]. However, these methods are quite expensive in runtime. A combined approach, which processes linear Gaussian parameters analytically and uses a numerical technique only for nonlinear and non-Gaussian parameters, was presented in [51]. The first-order canonical form was generalized to include non-Gaussian and nonlinear parameters, and a statistical approximation for the maximum of two generalized canonical forms was derived similarly as in the linear Gaussian case: first, a linear approximation using tightness probabilities as weighting factors is derived; then, the exact mean and variance values of the maximum of two generalized forms is computed. The first-order canonical form is generalized as

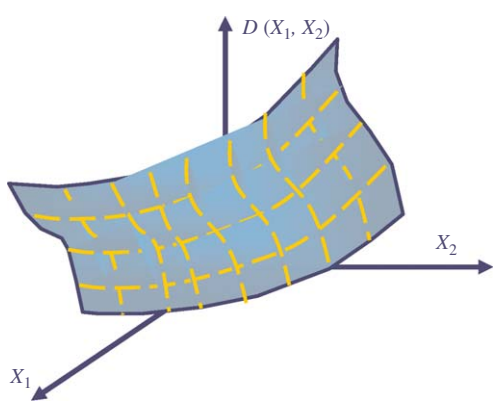

Fig. 21. Graphical representation of $D = D(X_1, X_2)$.

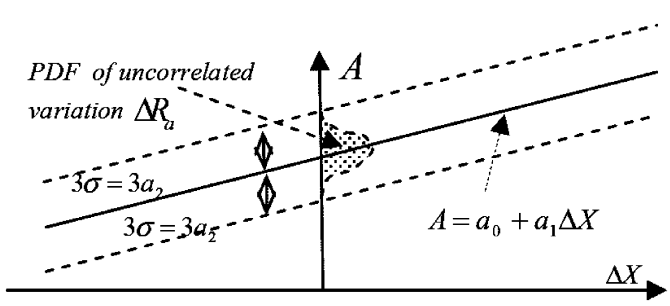

Fig. 22. Graphical representation of canonical form $A = a_0 + a_1\Delta X_1 + a_2\Delta R_a$ [51].

$$A = a_0 + \sum_{i=1}^{n_{LG}} a_{LG,i}\Delta X_{LG,i} + f_A(\Delta X_N) + a_{n_{LG}+1}\Delta R_a, \tag{26}$$

where $\Delta X_{LG,i}$ are linear Gaussian parameters and $a_{LG,i}$ their sensitivities, n_{LG} is the number of linear Gaussian parameters, $\Delta X_N = (\Delta X_{N,1}, \Delta X_{N,2,\ldots})$ is a vector of non-Gaussian and/or non-linear parameters, f_A is a function describing the dependence on non-Gaussian/nonlinear parameters (it should have zero mean value), and ΔR_a is a normalized Gaussian parameter for uncorrelated variation with its sensitivity $a_{n_{LG}+1}$. The generalization of the first-order canonical form (26) differs from the original one (3) only by the term $f_A(\Delta X_N)$ that describes dependencies of A on nonlinear and non-Gaussian parameters. For numerical computations, function f_A, which can be of arbitrary form, is represented by a table. Furthermore, there are no restrictions on the distribution of the non-Gaussian parameters that can be mutually correlated by means of a JPDF $\rho(\Delta X_{N,1}, \Delta X_{N,2,\ldots})$ specified by a table for numerical computation.

Propagation of arrival time in generalized canonical form through a timing edge with delay in the same form is similar to the pure linear Gaussian case. The only difference is the summation of nonlinear functions of the arrival time and delay, which can be performed numerically by summing tables describing these nonlinear functions. Hence, the sum of two generalized canonical forms is also a generalized canonical form. The computation of the sum of two timing quantities expressed as in (26), i.e., $C = sum(A, B)$, is expressed as in the following equation:

$$C = (a_0 + b_0) + \sum_{i=1}^{n_{LG}} (a_{LG,i} + b_{LG,i})\Delta X_{LG,i} + (f_A(\Delta X_N) + f_B(\Delta X_N)) + \sqrt{a_{n_{LG}+1}^2 + b_{n_{LG}+1}^2}\,\Delta R_c.$$

The approximation of the *max* of two generalized canonical forms is based on the same concept of tightness probability and computational approach as for the linear Gaussian case [37], so that the correlation of delays or arrival times is preserved. The parameters of the canonical form C_{appr} approximating the maximum of two generalized canonical forms A and B are obtained by the formulas

$$\begin{aligned} c_0 &= E[\max(A, B)], \\ c_i &= T_A a_i + (1 - T_A) b_i, \quad i = 1, \ldots, n_{LG}, \\ f_C(\Delta X_N) &= T_A f_A(\Delta X_N) + (1 - T_A) f_B(\Delta X_N), \end{aligned} \tag{27}$$

where T_A is the tightness probability. The sensitivity coefficient $c_{n_{LG}+1}$ to uncorrelated variation is computed to make the standard deviation of the approximation C_{appr} equal to the standard

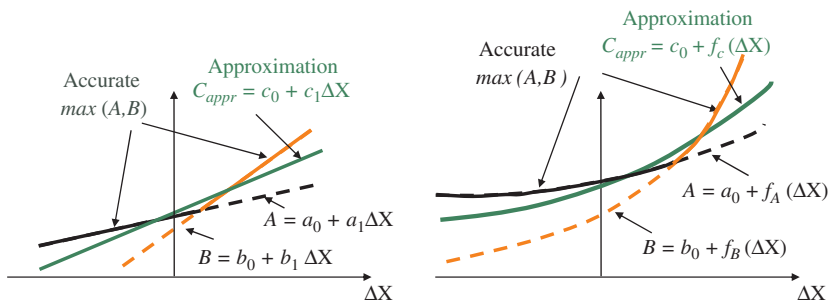

Fig. 23. Linear approximation of max of two canonical forms (left) and two generalized canonical forms (right) [51].

deviation of the exact maximum $C = \max(A, B)$. Similarly to the linear Gaussian case, the approximation of the maximum of two generalized canonical forms is linear: the coefficients c_i and function f_C are computed as linear combinations of coefficients a_i and b_i, and functions f_A and f_B, respectively, as in (27). Fig. 23 shows the linear approximation of the maximum of: (1) two canonical forms that depend only on one linear parameter (left); (2) two generalized canonical forms that depend only on one nonlinear parameter (right). The approximation of the maximum C_{appr} is represented by the green curve. The approximation of the maximum of two generalized canonical forms requires the computation of the tightness probability T_A, the mean, and the second moment of $\max(A, B)$. Considering the nonlinear and non-Gaussian parameter variations fixed, the expression for the generalized canonical form can be rewritten by combining the mean value a_0 and the term $f_A(\Delta X_N)$

$$A = (a_0 + f_A(\Delta X_N)) + \sum_{i=1}^{n_{LG}} a_{LG,i}\Delta X_{LG,i} + a_{n_{LG}+1}\Delta R_a. \tag{28}$$

Expression (28) can be considered as a canonical form A_{cond} with a mean value a_0+$f_A(\Delta X_N)$ and linear Gaussian parameters. All the sensitivities are the same as in the original generalized canonical form (26). If two generalized canonical forms A and B are represented as in (28), the conditional tightness probability, conditional mean, and second moments of $\max(A, B)$ are functions of the nonlinear and non-Gaussian parameters ΔX_N (with fixed values) given by

$$T_{A,cond}(\Delta X_N) = \mathrm{Prob}(A > B|\Delta X_N),$$
$$c_{0,cond}(\Delta X_N) = E[\max(A,B)|\Delta X_N],$$
$$m_{2,cond}(\Delta X_N) = E[(\max(A,B))^2|\Delta X_N].$$

The linear Gaussian parameters are independent of the nonlinear and non-Gaussian ones. Therefore, the joint conditional PDF of the linear Gaussian parameters at the condition of frozen values of nonlinear and non-Gaussian parameters is simply a JPDF of the linear Gaussian parameters. Hence, the same approach presented in [37] and reported in Section 4.2 can be used to compute the conditional tightness probability, mean, and second moments for the maximum of two generalized canonical forms at the condition that all nonlinear and non-Gaussian parameters are frozen, by substituting a_0+$f_A(\Delta X_N)$ and b_0+$f_B(\Delta X_N)$ for a_0 and b_0, respectively. The unconditional tightness probability, mean, and second moment of $\max(A, B)$ can be computed by integrating the conditional tightness probability, mean, and second moment over the space of nonlinear and non-Gaussian parameters with their JPDF, where such integration can be implemented by any numerical technique. Although the computational complexity for numerical integration by discretizing the integration region is exponential with respect to the number of nonlinear and non-Gaussian parameters, the experimental results presented in [51] show that for achieving a reasonable accuracy 5–7 discrete points of each variable are sufficient. This approach is practical for cases with up to 7–8 nonlinear and non-Gaussian variables. For higher dimensions the integrals can be computed by MC integration, and the overall approach rapidly becomes computationally expensive. Moreover, the approach [51] does not provide a solution in the presence of correlated non-Gaussian parameter distributions. Since the deviation from a normal distribution becomes more significant when the non-Gaussian random variables exhibit correlation, it is crucial to accurately manage the case where the non-Gaussian parameters may be correlated.

The work in [52] proposes a parameterized block-based SSTA algorithm that can handle both spatially correlated non-Gaussian as well as Gaussian distributions. The correlations are described using a grid structure similar to [38], which incorporates also non-Gaussian distributions. This approach works even for cases when the closed-form expression of the PDF of the sources of variation is not available, and it only requires the moments of the process parameter distributions. These moments are relatively easier to calculate from the process data files than the actual PDFs, and the procedure is based on a moment matching technique to generate the PDFs of the arrival time and delay variables.

To incorporate the effects of both Gaussian and non-Gaussian parameters in the SSTA framework presented in [52], all delays and arrival times are represented in linear form as

$$D = \mu + \sum_{i=1}^{n} b_i \cdot X_i + \sum_{j=1}^{m} c_j \cdot Y_j + e \cdot Z = \mu + \mathbf{B}^{\mathrm{T}} \cdot \mathbf{X} + \mathbf{C}^{\mathrm{T}} \cdot \mathbf{Y} + e \cdot Z, \tag{29}$$

where D is the random variable corresponding to a timing quantity (gate delay or arrival time at the input pin of a gate), X_i [Y_j] is a non-Gaussian [Gaussian] random variable corresponding to the physical parameter variation, b_i [c_j] is the first-order (linear) sensitivity of the timing quantity with respect to the ith non-Gaussian [jth Gaussian] parameter, Z is the uncorrelated parameter that could be either a Gaussian or non-Gaussian random variable, e is the sensitivity with respect to the uncorrelated variable, and n [m] is the number of correlated non-Gaussian [Gaussian] random variables. In the vector form, $\mathbf{B}$ and $\mathbf{C}$ are the sensitivity vectors for $\mathbf{X}$, the random vector of non-Gaussian parameter variations, and $\mathbf{Y}$, the random vector of Gaussian random variables, respectively. Gaussian and non-Gaussian parameters are statistical independent. The mean μ is adjusted so that $\mathbf{X}$ and $\mathbf{Y}$ are centered, i.e., each X_i, Y_j, and Z has zero-mean.

For computational and conceptual simplicity, it is useful to work with a set of statistically independent random variables. Since the random vector $\mathbf{Y}$ consists of correlated Gaussian random variables, a PCA transformation $\mathbf{R} = \mathbf{P}_Y\mathbf{Y}$ guarantees statistical independence for the components of the transformed vector $\mathbf{R}$ (for a Gaussian distribution, uncorrelatedness implies statistical independence). Such a property does not hold for general non-Gaussian parameters $\mathbf{X}$.

Independent component analysis (ICA) is a mathematical technique that accomplishes the desired goal of transforming a set of non-Gaussian correlated random variables into a set of random variables that are statistically as independent as possible, via a linear transformation. The approach described in [52] uses ICA as a preprocessing step to transform the correlated set of non-Gaussian random variables $X_1, \ldots, X_n$ to a set of statistically independent variables $S_1, \ldots, S_n$ by the following relation:

$$\mathbf{S} = \mathbf{W} \cdot \mathbf{X} \quad \text{where } S_i = \mathbf{W}_i^{\mathrm{T}} \cdot \mathbf{X} = \sum_{j=1}^{n} w_{ij} \cdot X_j \quad \forall i = 1, \ldots, n.$$

As in [38], the chip area is first tiled into a grid, and the covariance matrix associated with the random vector $\mathbf{X}$ is determined. Using the covariance matrix, and the underlying probability distributions of the variables in $\mathbf{X}$, samples of the correlated non-Gaussian variables are generated and are given as input to the ICA procedure, which produces as output the estimates of the matrix $\mathbf{W}$ and its inverse $\mathbf{A}$, called mixing matrix. For a specific grid, the independent components of the non-Gaussian random variables must be computed only once, and this can be carried out as a pre-characterization step. Hence, ICA does not have to be recomputed for different circuits or different placements of the same circuit, and this preprocessing step does not impact the runtime of the SSTA procedure. ICA is applied to the non-Gaussian parameters $\mathbf{X}$ and PCA to the Gaussian variables $\mathbf{Y}$, to obtain a set of statistically independent non-Gaussian variables $\mathbf{S}$ and a set of independent Gaussian variables $\mathbf{R}$. By substituting the respective transformation matrices $\mathbf{A}$ and $\mathbf{P}_Y$ in (29), the following *canonical delay model* can be derived:

$$D = \mu + \mathbf{B}'^{\mathrm{T}} \cdot \mathbf{S} + \mathbf{C}'^{\mathrm{T}} \cdot \mathbf{R} + e \cdot Z = \mu + \sum_{i=1}^{n} b_i' \cdot S_i + \sum_{j=1}^{m} c_j' \cdot R_j + e \cdot Z$$

$$\mathbf{B}'^{\mathrm{T}} = \mathbf{B}^{\mathrm{T}} \cdot \mathbf{A} \quad \mathbf{C}'^{\mathrm{T}} = \mathbf{C}^{\mathrm{T}} \cdot \mathbf{P}_Y^{-1}, \tag{30}$$

where $\mathbf{B}'^{\mathrm{T}}$ and $\mathbf{C}'^{\mathrm{T}}$ are the new sensitivity vectors with respect to the statistically independent non-Gaussian components $S_1, \ldots, S_n$ and Gaussian principal components $R_1, \ldots, R_m$. The inputs required for the SSTA approach in [52] are the moments of the random vector $\mathbf{X}$: $m_k(X_i) = E[X_i^k]$, which can be computed from mathematical tables if a closed-form PDF for the process parameters X_i is available, or from the process files. After performing ICA, the next step is to determine the moments of the independent components $S_1, \ldots, S_n$ from the moments of the correlated non-Gaussian parameters $m_k(X_i)$. The moments $E[S_i^k]$ can be used to compute the PDF (CDF) of any random delay variable expressed in the canonical form (30) using the binomial moment evaluation procedure proposed in [53], since this canonical form satisfies the independence requirement by construction. After computing the PDF and CDF of the delay and arrival time random variables expressed as linear canonical forms, the *sum* and *max* atomic operations of block-based SSTA can be performed to obtain a result in canonical form.

4.6. Quadratic timing modeling

In order to accurately account the impact of non-Gaussian and nonlinear parameters, most of the recent papers proposed as a solution quadratic timing models. In [54] it was reported that a quadratic delay model matches the MC simulations quite well. Moreover, for any Gaussian random variable, the skew (third-order moment) is always zero; hence, and non-zero skew distributions cannot be represented in linear delay models. In contrast, under nonlinear delay models, non-zero skews can be expressed by quadratic terms.

A quadratic timing model was proposed in [50] to capture the nonlinearity of the dependency of gate and wire delays as well as arrival times on the variation sources. In [50], the first-order canonical model was extended with second-order terms:

$$D = m + \alpha R + \sum_i b_i X_i + \sum_{i,j} a_{ij} X_i X_j, \tag{31}$$

where a_{ij} are quadratic coefficients and m is a constant term that in general might be different from the mean value of the delay timing variable. The difference with respect to the generalized canonical form (26) proposed in [51] is that in (26) the nonlinear/non-Gaussian parameters are represented by the nonlinear function $f_A(\Delta X_N)$, while in (31) they are characterized by the quadratic terms. The quadratic gate delay model is formulated by the second-order Taylor expansion with respect to the global sources of variation (evaluated around their mean value):

$$D_g \approx m_g + \alpha R + \frac{\partial D_g}{\partial L} L + \frac{\partial D_g}{\partial V} V + \frac{1}{2} \frac{\partial^2 D_g}{\partial L^2} L^2 + \frac{1}{2} \frac{\partial^2 D_g}{\partial V^2} V^2 + \frac{\partial^2 D_g}{\partial L \partial V} LV + \cdots, \tag{32}$$

where the coefficients in this Taylor expansion are computed during cell characterization, and are the same coefficients b_i and a_{ij} in (31)

$$b_i = \frac{\partial D_g}{\partial X_i}, \quad a_{ij} = \frac{1}{2} \frac{\partial^2 D_g}{\partial X_i \partial X_j}. \tag{33}$$

Assuming there are p global sources of variation, the Gaussian variation vector is defined as $\mathbf{X}_g = [X_1, X_2, \ldots, X_p]^{\mathrm{T}} \sim N(\mathbf{0}, \mathbf{\Sigma}_g)$. The correlation matrix $\mathbf{\Sigma}_g = E[\mathbf{X}_g \cdot \mathbf{X}_g^{\mathrm{T}}]$ in general is not a unit matrix, as these global variation random variables may be correlated. Eqs. (31) and (32) can be compacted into a quadratic form:

$$D_g = m_g + \alpha \cdot R + \mathbf{B}_g^{\mathrm{T}} \cdot \mathbf{X}_g + \mathbf{X}_g^{\mathrm{T}} \cdot \mathbf{A}_g \cdot \mathbf{X}_g, \tag{34}$$

where vector $\mathbf{B}_g$ and matrix $\mathbf{A}_g$ are a vectorized representation of the Taylor expansion coefficients (33). Similarly to the work [38], also in [50] the wire delay is expressed by the Elmore's delay model:

$$D_w = \sum_{i=1}^{N} \sum_{j=i}^{N} R_i \cdot C_j = \sum_{i=1}^{N} \sum_{j=i}^{N} \frac{r_s \cdot l^2 \cdot (c_s \cdot W_j + c_f \cdot T_j)}{W_i \cdot T_i}, \tag{35}$$

where R_i and C_i are the resistance and capacitance of the ith wire segment, r_s is the wire resistivity, c_s and c_f are the wire sheet and fringing capacitance, W_i and T_i are the width and thickness of the ith wire segment, and N is the number of wire segments with equal length l. Truncating the Taylor's expansion of (35) at the second order, the quadratic wire delay model can be expressed in compact form similarly to (34):

$$D_w = m_w + \alpha \cdot R + \mathbf{B}_w^{\mathrm{T}} \cdot \mathbf{X}_w + \mathbf{X}_w^{\mathrm{T}} \cdot \mathbf{A}_w \cdot \mathbf{X}_w, \tag{36}$$

where $\mathbf{X}_w$ is a $2N \times 1$ global variation vector: $\mathbf{X}_w = [W_1', W_2', \ldots, W_N', T_1', T_2', \ldots, T_N']^{\mathrm{T}} \sim N(\mathbf{0}, \mathbf{\Sigma}_w)$, while $W_i' = W_i - E[W_i]$ and $T_i' = T_i - E[T_i]$ are random variables, which in general are not statistically independent to each other since interconnects usually span a long distance and these variables may be spatially correlated. Due to the nonlinearity of the wire delay with respect to the process variations of width and thickness shown in Eq. (35), the delay distribution of the wire will not be Gaussian even if the width and thickness are usually considered to be Gaussian [5].

If there are q gate/wire delays in the input cone of the arrival time D_a and there are p global sources of variation impacting the q gate/wire delays, the arrival time will be approximated by the following quadratic form:

$$D_a = m_a + \boldsymbol{\alpha}_a^{\mathrm{T}} \cdot \mathbf{R}_a + \mathbf{B}_a^{\mathrm{T}} \cdot \mathbf{X}_a + \mathbf{X}_a^{\mathrm{T}} \cdot \mathbf{A}_a \cdot \mathbf{X}_a, \tag{37}$$

where random variation vectors $\mathbf{R}_a = [R_1, R_2, \ldots, R_q]^{\mathrm{T}} \sim N(\mathbf{0}, \mathbf{I})$ and $\mathbf{X}_a = [X_1, X_2, \ldots, X_p]^{\mathrm{T}} \sim N(\mathbf{0}, \mathbf{\Sigma}_a)$ are mutually independent local and global variations. If every arrival time in a circuit is approximated as a linear combination of its input gate/wire delays and all gate/wire delays have the quadratic delay form (34) and (36), then all timing variables in the circuit, including gate/wire delays and arrival times, will have the quadratic timing model:

$$D \sim Q(m, \boldsymbol{\alpha}, \mathbf{B}, \mathbf{A}) = m + \boldsymbol{\alpha}^{\mathrm{T}} \cdot \mathbf{R} + \mathbf{B}^{\mathrm{T}} \cdot \mathbf{X} + \mathbf{X}^{\mathrm{T}} \cdot \mathbf{A} \cdot \mathbf{X}. \tag{38}$$

In [50] it was demonstrated that for a quadratic timing quantity expressed as (38), its mean and variance are given by

$$\mu_D = E[D] = m + tr\{\mathbf{\Sigma} \cdot \mathbf{A}\},$$
$$\sigma_D^2 = \boldsymbol{\alpha}^{\mathrm{T}} \cdot \boldsymbol{\alpha} + \mathbf{B}^{\mathrm{T}} \cdot \mathbf{\Sigma} \cdot \mathbf{B} + 2 \cdot tr\{\mathbf{\Sigma}^2 \cdot \mathbf{A}^2\},$$

where $tr\{\cdot\}$ means trace and equals the sum of the diagonal elements of the matrix. The distribution of the quadratic delay model (38) can be computed by means of its characteristic function, analytically derived in [50].

If random variables X and Y are both expressed in quadratic form (38), the output of the *sum* operator is given by

$$Z = X + Y \sim Z(m_Z, \boldsymbol{\alpha}_Z, \mathbf{B}_Z, \mathbf{A}_Z),$$
$$m_Z = m_X + m_Y, \quad \boldsymbol{\alpha}_Z = \boldsymbol{\alpha}_X + \boldsymbol{\alpha}_Y,$$
$$\mathbf{B}_Z = \mathbf{B}_X + \mathbf{B}_Y, \quad \mathbf{A}_Z = \mathbf{A}_X + \mathbf{A}_Y.$$

In contrast, the *max* operator is intrinsically nonlinear, and it is necessary to evaluate if it can be approximated with a linear operator. The linearity of the *max* operator can be evaluated by the Gaussianity of the *max* output assuming the inputs are Gaussian. Skewness, which is a symmetry indicator of the distribution, can then be applied for the purpose of Gaussianity checking since a Gaussian distribution will always be symmetric. To propagate the quadratic timing model through the *max* operator, in [50] the *max* operation is first performed on two Gaussian inputs whose mean and variance match what is computed from the quadratic timing model. Then, the equations given in [42] are used to compute the output skewness. If the skewness is smaller than a threshold, then the *max* operator can be approximated by a linear operator. Otherwise, both inputs are placed into a *max*-tuple (*Mt*), which is a collection of random variables waiting to be *max*ed. The actual *max* operation can be postponed, since the *sum* operation for a *max*-tuple can be simply done as

$$Mt\{X, Y\} + D = Mt\{X + D, Y + D\}$$

and the *max* operation between two *max*-tuples is the merge of two tuples together:

$$\max(Mt\{X, Y\}, Mt\{U, V\}) = Mt\{X, Y, U, V\}.$$

To maintain the size of the *max*-tuple as small as possible, the linearity of the *max* operation is constantly checked between any two members of the *max*-tuple: if their *max* output skewness is small enough, then the *max* operation is performed on the two variables. With such conditional linear *max* operation, it is possible to control the error of the linear approximation for *max* operator within an acceptable range.

When two quadratic random variables X and Y expressed as in (38) are maximized with a linear approximation $Z = a \cdot X + b \cdot Y + c$, the approximation parameters a, b, and c, are computed assuming X and Y are Gaussian and using the equations in [42]. Hence, the quadratic timing variable $Z \sim Q(m_Z, \boldsymbol{\alpha}_Z, \mathbf{B}_Z, \mathbf{A}_Z)$ can be obtained by the following expressions:

$$\boldsymbol{\alpha}_Z = a \cdot \boldsymbol{\alpha}_X + b \cdot \boldsymbol{\alpha}_Y, \quad m_Z = a \cdot m_X + b \cdot m_Y + c,$$
$$\mathbf{B}_Z = a \cdot \mathbf{B}_X + b \cdot \mathbf{B}_Y, \quad \mathbf{A}_Z = a \cdot \mathbf{A}_X + b \cdot \mathbf{A}_Y.$$

Compared with the SSTA method based on first-order canonical model, the extra computation complexity of the method based on quadratic timing model stems from updating the quadratic coefficient matrix $\mathbf{A}$ at every arrival time propagation step. The number of quadratic coefficients is limited by the number of global variations and is usually a constant. Updating matrix $\mathbf{A}$ will not increase the computation complexity since it only involves moment computation of quadratic timing variables which is not dependent on the circuit size. To sum up, the computation complexity of SSTA based on quadratic timing model will be the same as its canonical timing model correspondence. In [54] the timing quantities such as gate and wire delays, arrival times, slacks, etc., are represented in the following quadratic form:

$$Y = \mathbf{X}^{\mathrm{T}} \cdot \mathbf{A} \cdot \mathbf{X} + \mathbf{B}^{\mathrm{T}} \cdot \mathbf{X} + C,$$

where $\mathbf{X} = (X_1, X_2, \ldots, X_n)^{\mathrm{T}}$ is the independent process parameter vector with normalized Gaussian distributions $N(0, 1)$ derived from PCA, $\mathbf{A}$ is a symmetric $n \times n$ matrix that contains the coefficients of the second-order terms, while $\mathbf{B}^{\mathrm{T}}$ is a $1 \times n$ vector, whose components are coefficients of the first-order terms, and C is a scalar constant term. Therefore, the *sum* operation of two random variables Y_1 and Y_2 is straightforward:

$$\begin{aligned} Y_1 &= \mathbf{X}^{\mathrm{T}} \cdot \mathbf{A}_1 \cdot \mathbf{X} + \mathbf{B}_1^{\mathrm{T}} \cdot \mathbf{X} + C_1 \\ Y_2 &= \mathbf{X}^{\mathrm{T}} \cdot \mathbf{A}_2 \cdot \mathbf{X} + \mathbf{B}_2^{\mathrm{T}} \cdot \mathbf{X} + C_2 \\ Y &= \mathrm{sum}(Y_1, Y_2) = Y_1 + Y_2 = \mathbf{X}^{\mathrm{T}} \cdot (\mathbf{A}_1 + \mathbf{A}_2) \cdot \mathbf{X} \\ &\quad + (\mathbf{B}_1^{\mathrm{T}} + \mathbf{B}_2^{\mathrm{T}}) \cdot \mathbf{X} + C_1 + C_2. \end{aligned} \tag{39}$$

In order to simplify the *max* operation, the cross terms $X_i X_j$ in the quadratic expression:

$$\begin{aligned} \max(Y_1, Y_2) = Y_1 &+ \max(0, Y_2 - Y_1) = Y_1 \\ &+ \max(0, \mathbf{X}^{\mathrm{T}} \cdot (\mathbf{A}_2 - \mathbf{A}_1) \cdot \mathbf{X} + (\mathbf{B}_2^{\mathrm{T}} - \mathbf{B}_1^{\mathrm{T}}) \cdot \mathbf{X} \\ &+ C_2 - C_1) \end{aligned}$$

should be removed, where Y_1 and Y_2 are expressed by quadratic forms as in (39). $(\mathbf{A}_2 – \mathbf{A}_1)$ is a symmetric matrix, thus it can be factorized as: $\mathbf{P}^{\mathrm{T}} \cdot \mathbf{\Sigma} \cdot \mathrm{P}$, where $\mathbf{\Sigma}$ is a diagonal matrix composed by the eigenvalues of $(\mathbf{A}_2 – \mathbf{A}_1)$ and $\mathbf{P}$ is the corresponding eigenvector matrix. If $\mathbf{Z} = \mathbf{P} \cdot \mathbf{X}$ and $\mathbf{\Phi} = (\mathbf{B}_2^{\mathrm{T}} - \mathbf{B}_1^{\mathrm{T}}) \cdot \mathbf{P}^{\mathrm{T}}$, then we obtain the following expression:

$$\max(Y_1, Y_2) = Y_1 + \max(0, \mathbf{Z}^{\mathrm{T}} \cdot \mathbf{\Sigma} \cdot \mathbf{Z} + \mathbf{\Phi} \cdot \mathbf{Z} + C_2 - C_1),$$

which no longer includes cross terms in the max operation. Since X_i's are independent Gaussian random variables, then also Z_i's are Gaussian random variables. Moreover, since the eigenvectors $\mathbf{P}$ of a symmetric matrix $(\mathbf{A}_2 – \mathbf{A}_1)$ are orthonormal, Z_i's are also uncorrelated; hence, Z_i's are also independent [53]. Therefore, it is possible to map the original parameter base into a new base without cross terms, perform the *max* operation under the new base, and map the results back into the original base. Based on this orthogonalization procedure, the inputs of the *max* operation in the approach presented in [54] are quadratic functions of an independent normalized base $\mathbf{X} = (X_1, X_2, \ldots, X_n)^{\mathrm{T}}$ without cross terms, where all X_i's are normalized Gaussian random variables $N(0, 1)$. The quadratic approximation of the nonlinear *max* operation in [54] is performed by solving a system of equations obtained via moment matching technique. However, this approach requires expensive numerical integrations.

A novel technique to model the gate and interconnect delay was presented in [55], where the authors proposed a delay model representation using orthogonal polynomials, which allows to independently computing the coefficients of the *max* of two delay expansions instead of using moment matching technique as in [54]. Their approach is based on the *Polynomial Chaos* theory. A second-order stochastic process can be

represented as

$$f = \sum_{i=0}^{\infty} a_i \psi_i, \tag{40}$$

where the functions ψ_i's are the *orthonormal basis*, and depend on the random variables modeling the underlying process variations. If the process variations are modeled with Gaussian variables, the basis functions are Hermite polynomials. In practice, the series expansion in (40) is truncated to a finite number of terms. While for any general distribution of random variables and any arbitrary function f the coefficients α_i can be estimated with expensive numerical techniques such as MC or generalized quadrature methods, for some specific distribution such as Gaussian, Uniform, etc., and a smooth function f, the integral can be evaluated with very high accuracy using N+1-order Gaussian quadrature, where N is the order of the polynomial that accurately approximates f. In [55] this method is used to perform library characterization; since standard cell delay and output slew can be modeled accurately using a second-order expansion, a third-order Gaussian quadrature can be used to estimate the expansion coefficients.

The delay is first expressed as a multi-variate function of both the process variations (e.g., V_{tn}, V_{tp}, T_{ox}, L), load capacitance C_{eff}, and input slew S_{in}, thus treating all these variables as deterministic quantities. By denoting with $\vec{Z}$ the normalized variables within the range $[-1, 1]$, the delay deterministic model can be expressed as a second-order Chebyshev polynomial series in the variables $\vec{Z}$. The coefficients of the Chebyshev polynomial expansion are obtained from the third-order interpolation of Chebyshev zeros on the Smolyak grid, to ensure some optimality in convergence while reducing the number of interpolation points:

$$d(\vec{Z}) = \sum_{i=0}^{N} \alpha_i \psi_i(\vec{Z}) = \alpha_0 + \sum_{i=1}^{6} \alpha_i Z_i + \sum_{i=1}^{6} \alpha_{6+i}(2Z_i^2 - 1) + \cdots + \alpha_N Z_5 Z_6.$$

Subsequently, the delay deterministic model is projected onto a second-order Hermite polynomial basis in the process variables and input slew. The coefficients of the second-order Hermite polynomial expansion, which are functions of the load capacitance C_{eff}, can be readily obtained for various values of C_{eff} by using the Galerkin technique. As a result, the delay can be expressed as

$$d(\bar{\xi}) = \sum_{i=0}^{N} \alpha_i \psi_i(\bar{\xi}), \tag{41}$$

where $\bar{\xi}$ represents the normalized (zero mean, unit variance) process and slew variables. A similar approach is adopted for modeling the output slew.

Due to manufacturing variations, some gate parameters on a die are random variables. Moreover, for a particular die, these random variables are functions of the gate location on the die, and can be modeled as a stochastic process $p(\bar{x}, \theta)$, where $\bar{x} = (x, y)$ is the location on the die, and θ belongs to the space of manufactured outcomes. Ideally, for each parameter, there are as many random variables as the number of gates in a die. In order to reduce the number of random variables, in [55] it was proposed to represent the process $p(\bar{x}, \theta)$ using the Karhunen–Loéve expansion (KLE):

$$p(\bar{x}, \theta) = \sum_{n=1}^{\infty} \sqrt{\lambda_n} \xi_n(\theta) \phi_n(\bar{x}),$$

where $\{\xi_n(\theta)\}$ is a set of uncorrelated random variables, λ_n are the eigenvalues, and $\{\phi_n(\bar{x})\}$ are the orthonormal eigenfunctions of the covariance function $C(\bar{x}_1, \bar{x}_2)$. The delay expansion of each gate i is obtained in terms of a common set of random variables by substituting the KLE corresponding to each random parameter of gate i in its delay expansion d_i. Once delays of all gates are obtained, it is possible to perform SSTA to compute the circuit delay in terms of the common set of variables. To propagate the delay through the circuit, both the *sum* and the *max* operations must be defined for the proposed delay expression. Given two delay expansions $d_1 = \sum_{i=1}^{n} \alpha_i \psi_i(\bar{\xi})$ and $d_2 = \sum_{i=1}^{n} \beta_i \psi_i(\bar{\xi})$, their sum can be obtained as $d_1 + d_2 = \sum_{i=1}^{n} (\alpha_i + \beta_i) \psi_i(\bar{\xi})$. The computation of the *max* is based on an efficient dimensionality reduction technique, which uses the moment matching methods to obtain the coefficients of the *max* of two delay expansions. The computation of the *sum* and *max* can also be extended to non-Gaussian variables. Therefore, the proposed approach can be used to propagate linear expansions of non-Gaussian variables.

Another approach where gate delay and arrival time distributions were modeled as polynomials using a Taylor-series expansion on the underlying parameters was presented in [48], where the degree of the polynomial depends on the magnitude of the variations and the required level of accuracy. In this work, the gate delay is a function of location-dependent parameters that are mutually independent random variables. Suppose P, Q, and R are such parameters (although the approach is very general and can be easily extended to more parameters); hence, the gate delay can be expressed similarly to (24)) as

$$D = D(P, Q, R), \tag{42}$$

where D can be a nonlinear function, and even if the random variables P, Q, and R are Gaussian distributions, in general the delay distribution (42) will not be Gaussian. Each parameter can be represented as a linear combination of the underlying random components as in (17), using the spatial correlation model described in Section 4.3. Therefore, expression (42) becomes

$$D = D(P_1, P_2, P_3, P_4, Q_1, Q_2, Q_3, Q_4, R_1, R_2, R_3, R_4) \tag{43}$$

and for the sake of conciseness the random variables in (43) are represented with the following notation:

$$D = D(X_1, X_2, X_3, X_4, X_5, X_6, X_7, X_8, X_9, X_{10}, X_{11}, X_{12}), \tag{44}$$

where all the random variables X_i are independent with zero mean and finite variance. The Taylor-series expansion of (44) around the mean values yields:

$$D = D(0) + \sum_{k=1}^{12} \left(\frac{\partial D}{\partial X_k}\right)_{X_k=0} X_k + \frac{1}{2} \sum_{k=1}^{12} \left(\frac{\partial^2 D}{\partial X_k^2}\right)_{X_k=0} X_k^2 + \cdots, \tag{45}$$

where $D(0)$ is the nominal value for gate delay (44) when all X_k random variables assume their nominal value. Expression (45) is similar to the quadratic gate delay represented by (32), and the gate delay is modeled as a general polynomial in the global variables X_k. It is worth pointing out that in (45) there are 66 second-order cross terms in the form $X_i X_j$, with $i \neq j$:

$$D = c_1 X_1 + \cdots + c_{12} X_{12} + c_{13} X_1^2 + \cdots + c_{24} X_{12}^2 + \cdots \tag{46}$$

and consequently there are 91 terms in expression (46), which is the second-order truncation of the Taylor-series expansion (45). It can be observed that by increasing the degree of the approximating polynomial, the number of terms increase and the error in approximation reduces. Therefore, there is a trade-off between runtime of statistical timing analysis and its accuracy. This trade-off can be controlled by the degree of the polynomial (46). Moreover, since all timing quantities in the circuit share the same global variables X_i, this approach enables to effectively capturing the correlations between them, similarly to the works [37,38].

The result of the *sum* operation between arrival time at the gate input A_i and the gate delay D_i approximated as a polynomial

(46) in the same independent global parameters is also a polynomial in the same global parameters. Likewise expression (9) the coefficient of each term in the resulting polynomial is the sum of the coefficients of the corresponding terms in A_i and D_i:

$$D_i = poly(X_1, X_2, \ldots, X_{12}),$$
$$A_i = poly(X_1, X_2, \ldots, X_{12}),$$
$$A^i_{out} = A_i + D_i = poly(X_1, X_2, \ldots, X_{12}). \quad (47)$$

Hence, the *max* operation among n polynomials obtained with (47) is a polynomial in the same global random variables

$$A_{out} = \max(A^1_{out}, A^2_{out}, \ldots, A^n_{out}) = poly(X_1, X_2, \ldots, X_{12}). \quad (48)$$

In [48] a regression-based strategy is proposed to compute the *max* operation by performing least square fitting, trying to find the best polynomial approximating the degree of polynomial (48) with the smallest error. To approximate A_{out} with a degree-two (i.e., quadratic) polynomial, the coefficients of the approximating polynomial should yield the smallest error against the actual *max* operation result obtained on a set of sampling vectors for the parameter X_i's. The advantage of using regression stems from the generality to handle timing distributions of any nature (not only Gaussians). However, the computational complexity of this approach grows exponentially with the polynomial order. To achieve the accuracy obtained from using a higher-order polynomial as well as runtime that is comparable to SSTA with linear delay models, a scheme using linear-modeling-based SSTA to drive the polynomial (i.e., quadratic) SSTA was proposed in [48]. Although the quadratic polynomial can represent the PDF/CDF of gate delays and arrival times more accurately than linear modeling, the mean and variance of the distributions are captured with reasonable accuracy with first-order polynomials. Therefore, in [48] a second-order polynomial modeling technique driven by linear modeling (which has lower runtime) was derived. With this technique the work presented in [48] avoided the complexity of solving a large (i.e., quadratic) polynomial regression problem at each gate (during the *max* operation) in block-based SSTA by solving a smaller linear regression problem and then performing moment matching (first two moments).

However, the proposed techniques to handle nonlinear delay dependency and non-Gaussian variation sources suffer from some limitations. The approach [52] addressed the non-Gaussian variation sources, but it is still based on a linear delay model. The nonlinear effects were considered in [50] and [54]: these works proposed a quadratic delay model. However, to keep the complexity under control they assumed that all the sources of variation must be represented by a Gaussian distribution, even though the delay may not be Gaussian. In order to compute the *max* between two delays D_1 and D_2, [50] treated D_1 and D_2 as Gaussians to obtain the tightness probability, even if there is no justification why the tightness probability formula can be applied to non-Gaussian distributions. Instead, [54] proposed to compute the $D = \max(D_1, D_2)$ by means of moment matching techniques, which requires several expensive numerical integrations. The works in [48,51] handled both nonlinear and non-Gaussian effects simultaneously. The first one proposed to compute $D = \max(D_1, D_2)$ by a regression-based strategy, while the latter dealt with the *max* operation through the concept of tightness probability, computed by means of expensive numerical multi-dimensional integrations. As a result, such methods are not suitable to handle a large number of non-Gaussian random variables.

A novel SSTA technique that efficiently performs the *max* operation and simultaneously handles both the nonlinear dependency and non-Gaussian distributions, was proposed in [56]. The authors represented the timing quantities in the following quadratic form, called *general canonical form*:

$$D = d_0 + \sum(a_i X_i + b_i X_i^2) + a_r X_r + b_r X_r^2, \quad (49)$$

where X_i are the global sources of variation, and X_r is the independent random variation. The X_i random variables may have arbitrary distributions with bounded values; they are assumed independent (if they are correlated, techniques like ICA [52] may be used to generate a new set of independent components) and centered with zero mean. To propagate the delay in block-based SSTA, not only it is necessary to efficiently compute the *sum* and *max* operations, but the timing results after each operation must be represented in the same general canonical form. Therefore, given D_1 and D_2 in the form (49)

$$D_1 = d_{01} + \sum(a_{i1} X_i + b_{i1} X_i^2) + a_{r1} X_{r1} + b_{r1} X_{r1}^2,$$
$$D_2 = d_{02} + \sum(a_{i2} X_i + b_{i2} X_i^2) + a_{r2} X_{r2} + b_{r2} X_{r2}^2$$

both $D = D_1 + D_2$ and $D = \max(D_1 D_2)$ must be represented as in (49). Denote $\Delta D_1 = D_1 - \mu_1$ and $\Delta D_2 = D_2 - \mu_2$, where μ_1 and μ_2 are the mean values of D_1 and D_2, respectively. Since both D_1 and D_2 are timing quantities, their values are physically lower- and upper-bounded:

$$-l \leqslant \Delta D_1 \leqslant l, \quad -h \leqslant \Delta D_2 \leqslant h.$$

To compute the *max*, the work [56] proposed a six-step flow. The first step computes the JPDF of D_1 and D_2, denoted as $g(v_1, v_2)$. If the JPDF of ΔD_1 and ΔD_2 is $f(v_1, v_2)$, it is easy to show that:

$$g(v_1, v_2) = f(v_1 - \mu_1, v_2 - \mu_2).$$

Then, the JPDF $f(v_1, v_2)$ is approximated by means of K-order Fourier series

$$f(v_1, v_2) \approx \sum_{p,q=-K}^{K} \alpha_{pq} e^{\zeta_p v_1 + \eta_q v_2}, \quad (50)$$

where $\zeta_p = jp\pi/l$ and $\eta_q = jq\pi/h$. In [56] an effective solution to simplify the computation of the Fourier coefficients α_{pq} was developed: for an arbitrary source of variation X_i, the X_i's range is divided into M small sub-regions, $S_1, \ldots, S_M$. Then, the Fourier transform of the PDF of X_i, denoted as $g_i(x_i)$, is pre-calculated for all pre-determined sub-regions of the variation source X_i, and the results are stored into a 1D lookup table. The valid region of each variation source is uniformly divided into twelve sub-regions and the fourth-order Fourier series is considered to represent the JPDF. In the second step, the raw moments $M_t = E[\max(D_1, D_2)^t]$ for $D = \max(D_1, D_2)$ are computed. According to (50), M_t can be written as

$$M_t = \sum_{p,q=-K}^{K} \alpha_{pq} L(t, p, q, l, h, \mu_1, \mu_2), \quad (51)$$

where $L = (t, p, q, l, h, \mu_1, \mu_2)$ can be efficiently evaluated with closed-form formulas. In the third step, the expectation $Ec_{i,t} = E[X_i^t \max(D_1, D_2)]$ is evaluated, by first obtaining the JPDF of X_i, ΔD_1, and ΔD_2, and then by computing $Ec_{i,t}$, similarly to the derivation of (51).

Finally, the last three steps are needed to reconstruct $D = \max(D_1, D_2)$ into the general canonical form (49), by first computing the coefficients a_i and b_i by matching $E[X_i^t \cdot \max(D_1, D_2)]$ for $t = 1, 2$; then by computing a_r and b_r in (49) by matching the second- and third-order moments of $\max(D_1, D_2)$; finally by computing d_0 in (49) by matching the first-order moment of $\max(D_1, D_2)$. The computation of $D = D_1 + D_2$ in the general canonical form (49) is straightforward, both for the nominal and global random variable coefficients, as they can be obtained by adding up the corresponding terms:

$$d_0 = d_{01} + d_{02}, \quad a_i = a_{i1} + a_{i2}, \quad b_i = b_{i1} + b_{i2}.$$

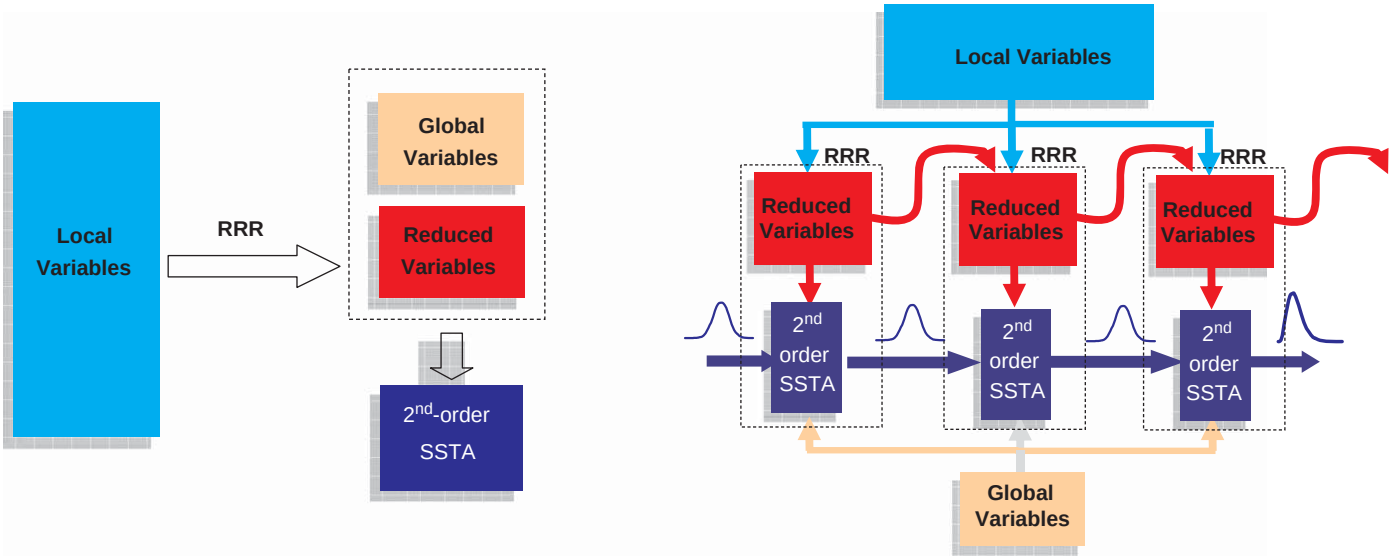

Fig. 24. Application of RRR technique to second-order SSTA: conceptual (on the left) and actual (on the right) [57].

For the uncorrelated random variable, there are two possible approaches. The first one is to keep the correlation between the addition result with the two input uncorrelated random variables X_{r1} and X_{r2}. The downside of this approach is that it causes the length of the general canonical form to grow longer after each addition. The second approach is to combine the two input uncorrelated random variables by matching both the second- and third-order central moments of the exact addition operation. The drawback of this method is that the correlation between D and X_{r1} and X_{r2} is lost. Since the two approaches complement each other, in [56] it was proposed to choose the first one when the coefficients of X_{r1} and X_{r2} are larger than a predefined threshold so that the correlation is not lost, and to choose the second technique when the coefficients of X_{r1} and X_{r2} are small so that the form can be kept compact.

An alternative approach to handle the increasing number of random variations, while maintaining the efficiency, was proposed in [57]. The approach is based on the linear reduced rank regression (RRR) that allows a powerful parameter reduction while considering the interdependency between parameters and the performances that depend on them. The conceptual application of RRR under the context of second-order SSTA is shown in Fig. 24 (left). For each circuit partitioning, RRR-based parameter reduction is performed once to reduce the number of local process variations and then a second-order SSTA can be carried out much more efficiently based on the original set of global variations and a reduced set of local variations. The way in which RRR is combined with SSTA is illustrated in Fig. 24 (right): RRR-based parameter reduction is intertwined with each SSTA processing step to dynamically control the parameter dimension. For strong second-order effects, the linear RRR framework can be extended to generate a nonlinear RRR regression model [58]. Results reported in [57] demonstrate that the additional cost of the RRR-based parameter reduction algorithm can be almost neglected when compared to the complexity of the second-order SSTA algorithm.

4.7. Statistical static timing analysis including crosstalk effects

Along with process variations, technology scaling to smaller dimensions also causes the dominant portion of wiring capacitance to be the inter-layer neighboring wire capacitance. Hence, the gate delay can be greatly impacted by the switching activity on neighboring wires. This change in delay due to capacitive coupling is referred to as *delay noise* and it contributes to a significant portion (up to 40% stage delay error [59]) of the circuit delay. In traditional STA, the problem of delay computation in the presence of crosstalk can be formulated as computing the earliest and the latest arrival time among all possible waveforms of the aligned aggressors. The timing window for a given circuit can be computed by means of iterative algorithms [59]. In each loop, the early and the late arrival times at the primary inputs are propagated to the primary outputs taking into account the influence of aggressor gates. The resulting timing window of each net is compared with its aggressors to decide the aligned aggressors. The aggressor whose timing windows are not overlapped with the victim net will be set as unaligned aggressor in the next loop to shrink the timing window. As shown in Fig. 25, the timing windows for two nets are overlapped if and only if

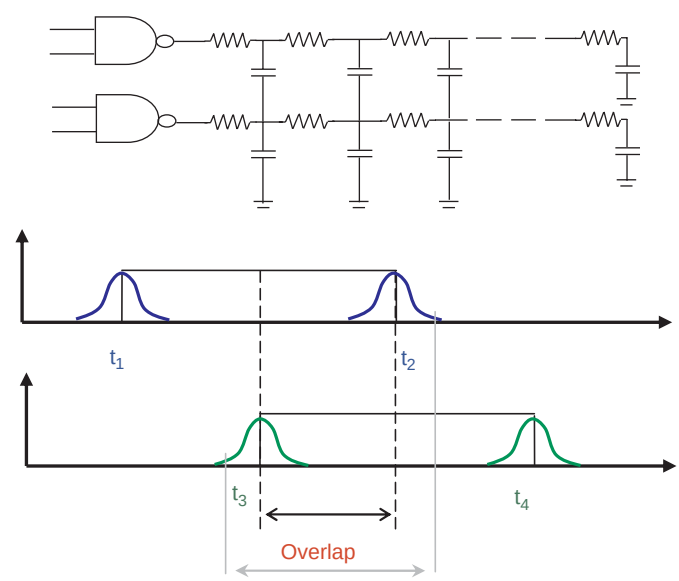

Fig. 25. Statistical Static Timing Analysis with coupling: capacitive-coupled interconnects with their driver gates (above) and arrival timing windows in presence of variability (below) [60].

$t_1 < t_4$ and $t_3 < t_2$, where t_1, t_3 are the early arrival times and t_2, t_4 are the late arrival times. Following this iterative timing window alignment procedure [60], proposed to extend SSTA to consider the impact of variations and coupling effects concurrently. In SSTA, the earliest arrival time and latest arrival time for a timing window of a given net become random variables. The overlap of two timing windows can no longer be simply determined by the condition $t_1 < t_4$ and $t_3 < t_2$, represented by the timing interval between the dashed lines in Fig. 25. On the other hand, since t_1, t_2, t_3, t_4 are all random variables, new random variables $(t_4 - t_1)$ and $(t_2 - t_3)$, along with the overlap condition can be defined as follows:

$$\mu_{t_4-t_1} + 3\sigma_{t_4-t_1} > 0,$$
$$\mu_{t_2-t_3} + 3\sigma_{t_2-t_3} > 0.$$

By using the 3σ values to determine the overlap of two timing windows, represented by the timing interval between the grey lines in Fig. 25, the proposed method prevents the over-shrink of the timing windows and preserves the earliest and latest arrival times. Furthermore, the correlations between different arrival times are inherently incorporated into the new random variables, thus removing any unnecessary pessimism in the timing window alignment. The mean value and the standard deviation for new random variable $t_i - t_j$ can be computed from the existing mean and covariance tables:

$$\mu_{t_i-t_j} = Mean(t_i) - Mean(t_j),$$
$$\sigma_{t_i-t_j} = \sqrt{Var(t_i) - 2Cov(t_i, t_j) + Var(t_j)}.$$

However, this approach estimates the switching window overlap deterministically, based on the worst-case values, leading to a pessimistic computation of the victim delay variation due to coupling. An alternative solution to statistical timing analysis with coupling is to consider a *distribution of switching windows* on each wire of the circuit [61]. This view is obtained by first considering the uncertainty from ignorance of functional information, and then the uncertainty from variability. Each window is denoted by a best and worst value. Therefore, the distribution of the switching windows contains the distributions of the best- and worst-case values. The two distributions thus obtained are represented as the distributions of two correlated random variables, respectively. The window formed using these random variables as the best and the worst-case value, respectively, contains all possible signal switching distributions. This transformation of the original solution gives an alternate view of the solution to SSTA with coupling as a *window of signal switching distributions*. This view of the solution can also be obtained by considering the uncertainty from variability first. Hence, SSTA on a wire i computes a distribution of a single signal switching. However, when the uncertainty from functional information and input configuration is also considered, the timing information on i is then a set of signal switching distributions, which is represented by a window of signal switching distributions. In [61] the concept of *statistical switching window* was introduced as a representation for a set of random variables. For any random variable x_i^* in the set, a lower and an upper bound on the probability that x_i^* is not less (or more) than a given real value c is considered. The statistical switching window extends the bounds over the entire range of c, in the form of two distribution of correlated random variables x_i^l and x_i^u, respectively. These two random variables are called the bounding Random Variables of the statistical switching window x_i. Mathematically, the statistical switching window is defined as follows:

$$x_i = [x_i^l, x_i^u] = \{x_i^* : \Pr(x_i^l \leqslant x_i^* \leqslant x_i^u) = 1\},$$

where $\Pr(k)$ denotes the probability of event k. Then, the *inclusion* relation between two statistical switching windows x_i and x_j is formally defined as

$$x_i \subseteq x_j def(\Pr(x_j^l \leqslant x_i^l \leqslant x_i^u \leqslant x_j^u) = 1).$$

Both a statistical switching window example and the inclusion relation between statistical switching windows are graphically illustrated in Fig. 26. The amount of delay noise is function of the overlap between the switching windows on the coupled nets. For two statistical switching windows x_i and x_j, the overlap between the windows is a random variable O_{ij} defined as follows:

$$O_{ij} = \min(x_i^u, x_j^u) - \min(x_i^l, x_j^l).$$

Since the overlap between two statistical switching windows is a random variable, the coupling induced delay noise is consequently a random variable.

To compute the worst-case delay on a wire i when coupling effects are considered, we need to add to the delay on the wire when coupling effects are not considered, and the coupling induced delay noise due to each wire it couples with. In [61], an example of computation of a delay noise as a random variable based on a simple coupling model is illustrated. The considered coupling model is given by

$$D = \begin{cases} D^O & overlap, \\ D^N & no\ overlap, \end{cases}$$

where D^O and D^N are the values assigned to D, depending on whether the statistical switching windows between the coupled

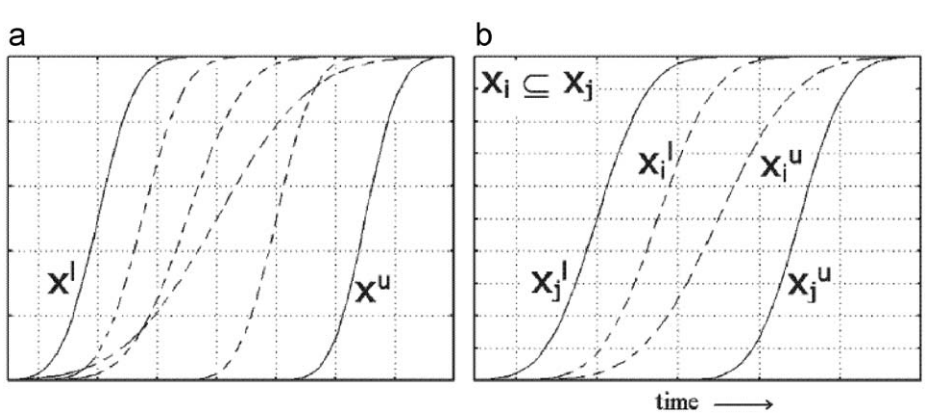

Fig. 26. A statistical switching window for a set of distributions (a) and the inclusion relation between two statistical switching windows (b) [61].

wires overlap or not, respectively. Using PCA, all random variables are expressed as a weighted sum of independent and orthonormal random variables ξ_i, $i = 1,2,\ldots,n$. The delay noise is a random variable and it is expressed as

$$D = d_0 + \sum_{i=1}^{n} d_i \xi_i = \begin{cases} d_0^O + \sum_{i=1}^{n} d_i^O \xi_i & \textit{overlap}, \\ d_0^N + \sum_{i=1}^{n} d_i^N \xi_i & \textit{no overlap}. \end{cases}$$

The computation of the d_i coefficients as a function of the overlap random variable O is reported in [61]. While the approach proposed in [61] is extensible to an arbitrary coupling model, it cannot use Gaussian switching windows because the assumption of a Gaussian distribution for the bounding Random Variables prohibits the generic use of the inclusion relation between the switching windows. In this case, the solution is to replace Gaussian distributions with *truncated* Gaussian distributions for representing the bounding Random Variables. Arithmetic operations on Gaussians are used identically for truncated Gaussians, although they involve some approximations.

Another approach to include crosstalk noise into SSTA was proposed in [62]. Given a quadratic model of the delay change curve *DCC* which captures the dependence of delay noise on the aggressor-victim input skew, graphically represented in Fig. 27, and an input skew distribution in canonical form, the proposed approach allows to obtaining closed-form expressions of the resulting delay noise distribution. Since the correlations in the input skew are preserved exactly in the delay noise distribution, it is possible to express the delay noise in canonical form. Without loss of generality, the input skew distribution is given by

$$s = s_0 + s_1 x_1 + s_2 x_2,$$

where s_0 is the mean, and s_1 and s_2 are the sensitivities with respect two independent standard normal random variables x_1 and x_2. Since the process parameters are Gaussians, the input skew PDF $f_s(s)$ is therefore normally distributed with mean μ and variance σ expressed by

$$\mu = s_0, \quad \sigma = \sqrt{s_1^2 + s_2^2}.$$

Supposing that the delay change curve *DCC* is piece-wise quadratic, as depicted in Fig. 27:

$$DCC = \begin{cases} 0, & s < z_0 \\ a_1 s^2 + b_1 s + c_1, & z_0 \leqslant s \leqslant z_1 \\ a_2 s^2 + b_2 s + c_2, & z_1 \leqslant s \leqslant z_2 \\ 0, & s > z_2 \end{cases}$$

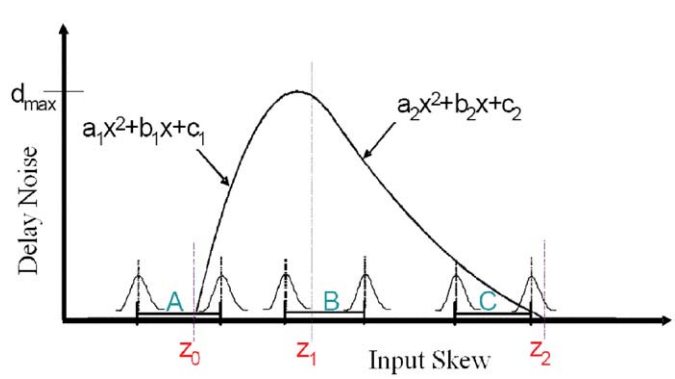

Fig. 27. Delay change curve captures the dependence of the delay noise on input skew [62].

then the PDF of the delay noise $f_d(s)$ as a function of the input skew distribution can be directly obtained by applying the basic theory of probability and statistics:

$$f_d(s) = \frac{f_s(r_1)}{|2a_1 r_1 + b_1|} + \frac{f_s(r_2)}{|2a_2 r_2 + b_2|}, \tag{52}$$

where r_1 and r_2 are the smaller and larger roots of the two quadratic pieces of the *DCC*. The delay noise in (52) is not necessarily Gaussian. However, using the PDF of delay distribution from (52), it is possible to compute the first and the second moment of delay noise in closed form. Therefore, the canonical form of delay noise can be constructed by matching the first two moments, while the correlations of the delay noise distribution are assigned by using the sensitivities of the given single aggressor-victim input skew distribution to process parameters. The proposed analytical technique can be extended so that the worst-case delay noise computation can be performed within the current SSTA framework with statistical timing windows, instead of single skew distribution. The solution is based on the result reported in [63], where it is shown that regardless of the aggressor transition, the worst-case delay noise occurs when the victim input transition occurs at the latest point in its timing window. Therefore, for computing the worst-case delay noise, in [62] only the distribution of late victim input arrival time was considered. Given the statistical timing window at the input of the aggressor, the early and late aggressor input arrival time distributions are subtracted from the late victim arrival time distribution to obtain the statistical skew window. The arrival time distributions of end points of the skew window are referred to as early and late skew distributions. As shown in Fig. 27, the skew window can align with the *DCC* in three different ways, denoted as Case A (when the mean of late skew distribution is less than the worst-case skew value if the *DCC*), Case B (when the mean of late skew distribution is less than z_1 and the mean of early skew distribution is less than z_1) and Case C (when the mean of early skew distribution is greater than z_1). Since any skew distribution which lies within the skew window is feasible, for Case B, the delay noise is modeled by its worst-case value d_{max}. Note that in Case A, the *DCC* is a monotonic function of skew. Therefore, the mean delay noise will be maximized when the mean of the feasible skew distribution coincides with the late skew distribution. Therefore, the delay noise distribution in canonical form can be analytically computed as a function of the late skew distribution. Similarly, for Case C, the early skew distribution can be used to obtain the delay noise distribution. As a result, given the statistical timing window from block-based SSTA, the delay noise distribution can be analytically computed. Since it is in canonical form, it can trivially be added to the late victim output arrival time distribution and propagated downstream.

5. Conclusions

As new silicon technologies keep shrinking the transistor size, it becomes more and more difficult to precisely control the process parameters during fabrication. As a consequence, both the number and the magnitude of independent sources of variations are increasing. These unavoidable process parameter fluctuations may significantly impact the design performance, often resulting in a considerable parametric yield loss. Therefore, the accurate prediction of the process variation impact on circuit performance is a critical issue.

Traditionally, the design performance evaluation in presence of variability has been performed either by running multiple STA at different process parameter "corners", or by verifying the design in the "worst-case" ("best-case") corner. Due to the growing

quantity of process parameters, the first approach may result in an unacceptable number of timing runs, while the latter may lead to overly pessimistic performance estimation.

SSTA is a promising solution to overcome these limitations. During SSTA, all the timing quantities such as delay, arrival time, and slack are treated as probability distributions. Therefore, the probability distribution of the circuit performance under parameter variability can be predicted in a single analysis. Moreover, SSTA may accurately account for the actual process parameter distributions and their correlations, thus potentially avoiding overly conservative design. Furthermore, SSTA has many other advantages over traditional STA. For instance, it may provide information about the design sensitivity to different process parameters, thus driving designers to implement a more robust design. Moreover, it may predict the parametric yield curve, thus allowing an early decision making on risk management.

Similarly to traditional STA tools, existing SSTA methods can be classified based on the different algorithmic approaches used to compute the delay distribution, i.e., path-based or block-based. Path-based analysis is accurate and can accurately capture correlations, but it suffers from important limitations: its complexity grows exponentially with respect to the circuit size; therefore, only a tiny fraction of the billions of paths typically present in modern SoCs can be analyzed. Moreover, the path-based algorithm is not incremental. On the other hand, the block-based approach has a computational complexity linear with the circuit size. By allowing the analysis to cover all possible paths simultaneously, it responds incrementally to timing queries after changes to the circuit are carried out. Furthermore, it provides the diagnostics necessary to improve the design robustness. Hence, even if the block-based algorithm is less precise and has some limitations considering the correlations, it is the engine that underlies industrial SSTA tools today.

Most of the recent research activity on SSTA has been devoted to mitigate the block-based approach limitations, i.e., the realistic correlation handling and accurate delay calculation. In this survey, the main approaches proposed in the literature addressing these challenges have been analyzed and discussed.

One of the most significant novelties is the *canonical first-order delay* model that allows to considering both global correlations and independent randomness. By means of this delay model, the global and the local criticality probabilities can be computed, which are useful for design diagnostics.

Along with the canonical first-order model, the concept of *tightness probability* has been introduced, providing an improvement to accurately and efficiently compute the *max* of *n* correlated Gaussian distributions, which was a fundamental source of error inherent with the block-based approach.

Several works have addressed the spatial correlation modeling, since ignoring such correlations may result in an underestimation of the variability impact. The main approaches, *quad-tree partitioning*, *grid-die partitioning*, and *grid-based radial* modeling have been analyzed and compared.

Most of the block-based approaches proposed in the past assumed that all parameters had normal Gaussian probability distributions and affected gate delays linearly. However, some process parameters have significantly non-Gaussian probability distributions. Moreover, as the process variations are becoming larger, the linear approximation is not accurate enough. Therefore, a few techniques have been proposed in the literature in order to include non-Gaussian and nonlinear parameters in SSTA. In this survey, the *generalized canonical delay form*, the *quadratic*, and the *polynomial timing delay* models have been discussed.

Finally, some techniques for including the impact of crosstalk noise into the statistical timing analysis algorithms have been described and compared.

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References

[1] D. Blaauw, K. Chopra, A. Srivastava, L. Scheffer, Statistical timing analysis: from basic principles to state of the art, IEEE Trans. Computer-Aided Design 27 (2008) 589–607.
[2] D. Pandini, G. Desoli, A. Cremonesi, Computing and design for software and silicon manufacturing, in: Proceedings of the International Conference on VLSI-SoC, October 2007, pp. 122–127.
[3] D. Pandini, Innovative design platforms for reliable SoCs in advanced nanometer technologies, in: Proceedings of the International On-Line Testing Symposium, July 2007, p. 254.
[4] S.R. Nassif, Modeling and forecasting of manufacturing variations, in: Proceedings of the ASP-DAC, February 2001, pp. 145–149.
[5] S.R. Nassif, Modeling and analysis of manufacturing variations, in: roceedings of the Custom Integrated Circuits Conference, May 2001, pp. 223–228.
[6] L. Stok, J. Koehl, Structured CAD: technology closure for modern ASICs, Tutorial, DATE, March 2004.
[7] A. Dharchoudhury, S.M. Kang, Worst-case analysis and optimization of VLSI circuit performances, IEEE Trans. Computer-Aided Design 14 (1995) 481–492.
[8] E. Acar, S. Nassif, Y. Liu, L.T. Pileggi, Assessment of true worst case circuit performance under interconnect parameter variations, in: Proceedings of the International Symposium on Quality Electronic Design, March 2001, pp. 431–436.
[9] C. Visweswariah, Death, taxes and failing chips, in: Proceedings of the Design Automation Conference, June 2003, pp. 343–347.
[10] S.G. Duvall, Statistical circuit modeling and optimization, in: Proceedings of the International Workshop on Statistical Metrology, June 2000, pp. 56–63.
[11] K.A. Bowman, S.G. Duvall, J.D. Meindl, Impact of die-to-die and within-die parameter fluctuations on the maximum clock frequency distribution for gigascale integration, IEEE J. Solid-State Circuits 37 (2002) 183–190.
[12] D. Boning, S.R. Nassif, Models of process variations in device and interconnect, in: A. Chandrakasan, W. Bowhill, F. Fox (Eds.), Design of High Performance Microprocessor Circuits, IEEE Press, New York, 2000.
[13] D.J. Frank, Y. Taur, M. Ieong, H.S.P. Wong, Monte Carlo modeling of threshold variation due to Dopant fluctuations, in: Proceedings of the VLSI Technology Symposium, June 1999, pp. 169–170.
[14] P.S. Zuchowski, P.A. Habitz, J.D. Hayes, J.H. Oppold, Process and environmental variation impacts on ASIC timing, in: Proceedings of the International Conference on Computer-Aided Design, November 2004, pp. 336–342.
[15] International Technology Roadmap for Semiconductors, 2003 edition, Semiconductor Industry Association, 2003.
[16] U. Schlichtmann, DFM/DFY design for manufacturability and yield—influence of process variations in digital, analog and mixed-signal circuit design, in: Proceedings of the DATE, March 2006, pp. 387–392.
[17] R.B. Hitchcock, Timing verification and the timing analysis program, in: Proceedings of the Design Automation Conference, June 1982, pp. 594–604.
[18] L.T. Pillage, R.A. Rohrer, Asymptotic waveform evaluation for timing analysis, IEEE Trans. Computer-Aided Design 9 (1990) 352–366.
[19] P. Feldman, R.W. Freund, Efficient linear circuit analysis by Pade approximation via the Lanczos process, IEEE Trans. Computer-Aided Design 14 (1995) 639–649.
[20] A. Odabasiouglu, M. Celik, L.T. Pileggi, PRIMA: passive reduced-order interconnect macromodeling algorithm, in: Proceedings of the International Conference on Computer-Aided Design, November 1997, pp. 58–65.
[21] S. H. C. Yen, D. C. Du, and S. Ghanta, Efficient algorithms for extracting the K most critical paths in timing analysis, in: Proceedings of the Design Automation Conference, June 1989, pp. 649–654.
[22] T. Kirkpatrick, N. Clark, PERT as an aid to logic design, IBM J. Res. Dev. 10 (1966) 135–141.
[23] B. Choi, D.M.H. Walker, Timing analysis of combinational circuits including capacitive coupling and statistical process variation, in: Proceedings of the VLSI Test Symposium, April 2000, pp. 49–54.
[24] S. Tasiran, A. Demir, Smart Monte Carlo for yield estimation, in: Proceedings of the International Workshop on Timing Issues, February 2006.
[25] R. Kanj, R. Joshi, S. Nassif, Mixture importance sampling and its application to the analysis of SRAM designs in the presence of rare failure events, in: Proceedings of the Design Automation Conference, June 2006, pp. 69–72.
[26] S.R. Naidu, Speeding up Monte Carlo simulation for statistical timing analysis of digital integrated circuits, in: Proceedings of the International Conference on VLSI Design, January 2007, pp. 265–270.
[27] V. Veetil, D. Sylvester, D. Blaauw, Criticality aware latin hypercube sampling for efficient statistical timing analysis, in: Proceedings of the International Workshop on Timing Issues, February 2007.
[28] A. Singhee, R.A. Ruthenbar, From finance to flip flop: a study of fast Quasi-Monte Carlo methods from computational finance applied to statistical circuit analysis, in: Proceedings of the International Symposium on Quality Electronic Design, March 2007, pp. 685–692.

[29] V. Veetil, D. Sylvester, D. Blaauw, Efficient Monte Carlo based incremental statistical timing analysis, in: Proceedings of the International Workshop on Timing Issues, February 2008.
[30] L. Scheffer, The count of Monte Carlo, in: Proceedings of the International Workshop on Timing Issues, February 2004.
[31] J.A.G. Jess, K. Kalafala, S.R. Naidu, R.H.J. Otten, C. Visweswariah, Statistical timing for parametric yield prediction of digital integrated circuits, in: Proceedings of the Design Automation Conference, June 2003, pp. 932–937.
[32] C.S. Amin, N. Menezes, K. Killpack, F. Dartu, U. Choudhury, N. Hakim, Y.I. Ismail, Statistical static timing analysis: how simple can we get?, in: Proceedings of the Design Automation Conference, June 2005, pp. 652–657.
[33] A. Devgan, C. Kashyap, Block-based static timing analysis with uncertainty, in: Proceedings of the International Conference on Computer-Aided Design, November 2003, pp. 607–614.
[34] M. Orshansky, A. Bandyopadhyay, Fast statistical timing analysis handling arbitrary delay correlations, in: Proceedings of the Design Automation Conference, June 2004, pp. 337–342.
[35] M. Orshansky, K. Keutzer, A general probabilistic framework for worst case timing analysis, in: Proceedings of the Design Automation Conference, June 2002, pp. 556–561.
[36] A. Agarwal, V. Zolotov, D. Blaauw, Statistical timing analysis using bounds and selective enumeration, IEEE Trans. Computer-Aided Design 22 (2003) 1243–1260.
[37] C. Visweswariah, K. Ravindran, K. Kalafala, S.G. Walker, S. Narayan, First-order incremental block-based statistical timing analysis, in: Proceedings of the Design Automation Conference, June 2004, pp. 331–336.
[38] H. Chang, S.S. Sapatnekar, Statistical timing analysis considering spatial correlations using a single PERT-like traversal, in: Proceedings of the International Conference on Computer-Aided Design, November 2003, pp. 621–625.
[39] M.R.C.M. Berkelaar, Statistical delay calculation: a linear time method, in: Proceedings of the International Workshop on Timing Issues, December 1997, pp. 15–24.
[40] E.T.A.F. Jacobs, M.R.C.M. Berkelaar, Gate sizing using a statistical delay model, in: Proceedings of the DATE, March 2000, pp. 283–290.
[41] S. Tsukiyama, M. Tanaka, M. Fukui, A new statistical static timing analyzer considering correlation between delays, in: Proceedings of the International Workshop on Timing Issues, December 2000, pp. 27–33.
[42] C.E. Clark, The greatest of a finite set of random variables, Oper. Res. 9 (1961) 145–162.
[43] D. Sinha, H. Zhou, N.V. Shenoy, Advances in computation of the maximum of a set of Gaussian random variables, IEEE Trans. Computer-Aided Design 26 (2007) 1522–1533.
[44] K. Chopra, B. Zhai, D. Blaauw, D. Sylvester, A new statistical max operation for propagating skewness in statistical timing analysis, in: Proceedings of the International Conference on Computer-Aided Design, November 2006, pp. 237–243.
[45] A. Agarwal, D. Blaauw, V. Zolotov, Statistical timing analysis for intra-die process variations with spatial correlations, in: Proceedings of the International Conference on Computer-Aided Design, November 2003, pp. 900–907.
[46] A. Agarwal, D. Blaauw, V. Zolotov, S. Sundareswaran, M. Zhou, K. Gala, R. Panda, Statistical delay computation considering spatial correlations, in: Proceedings of the ASP-DAC, January 2003, pp. 271–276.
[47] V. Mehrotra, S.L. Sam, D. Boning, A. Chandrakasan, R. Vallishayee, S. Nassif, A methodology for modeling the effects of systematic within-die interconnect and device variation on circuit performance, in: Proceedings of the Design Automation Conference, June 2000, pp. 172–175.
[48] V. Khandelwal, A. Srivastava, A general framework for accurate statistical timing analysis considering correlations, in: Proceedings of the Design Automation Conference, June 2005, pp. 89–94.
[49] B. Cline, K. Chopra, D. Blaauw, Y. Cao, Analysis and modeling of CD variation for statistical static timing, in: Proceedings of the International Conference on Computer-Aided Design, November 2006, pp. 60–66.
[50] L. Zhang, W. Chen, Y. Hu, J.A. Gubner, C.C.-P. Chen, Correlation-preserved non-Gaussian statistical timing analysis with quadratic timing model, in: Proceedings of the Design Automation Conference, June 2005, pp. 83–88.
[51] H. Chang, V. Zolotov, S. Narayan, C. Visweswariah, Parameterized block-based statistical timing analysis with non-Gaussian parameters, nonlinear delay functions, in: Proceedings of the Design Automation Conference, June 2005, pp. 71–76.
[52] J. Singh, S. Sapatnekar, Statistical timing analysis with correlated non-Gaussian parameters using independent component analysis, in: Proceedings of the Design Automation Conference, July 2006, pp. 155–160.
[53] X. Li, J. Le, P. Gopalakrishnan, L.T. Pileggi, Asymptotic probability extraction for non-normal distributions of circuit performance, in: Proceedings of the International Conference on Computer-Aided Design, November 2004, pp. 1–9.
[54] Y. Zhan, A.J. Strojwas, X. Li, L.T. Pileggi, D. Newmark, M. Sharma, Correlation-aware statistical timing analysis with non-gaussian delay distributions, in: Proceedings of the Design Automation Conference, June 2005, pp. 77–82.
[55] S. Bhardway, P. Ghanta, S. Vrudhula, A framework for statistical timing analysis using nonlinear delay and slew models, in: Proceedings of the International Conference on Computer-Aided Design, November 2006, pp. 225–230.
[56] L. Cheng, J. Xiong, L. He, Nonlinear statistical static timing analysis for non-Gaussian variation sources, in: Proceedings of the Design Automation Conference, June 2007, pp. 250–255.
[57] Z. Feng, P. Li, Y. Zhan, Fast second-order statistical static timing analysis using parameter dimension reduction, in: Proceedings of the Design Automation Conference, June 2007, pp. 244–249.
[58] Z. Feng, P. Li, Performance-oriented statistical parameter reduction of parameterized systems via reduced rank regression, in: Proceedings of the International Conference on Computer-Aided Design, November 2006, pp. 868–875.
[59] R. Arunachalam, K. Rajagopal, L.T. Pileggi, TACO: timing analysis with coupling, in: Proceedings of the Design Automation Conference, June 2000, pp. 266–269.
[60] J. Le, X. Li, L.T. Pileggi, STAC: statistical timing analysis with correlation, in: Proceedings of the Design Automation Conference, June 2004, pp. 343–348.
[61] D. Sinha, H. Zhou, Statistical timing analysis with coupling, IEEE Trans. Computer-Aided Design 25 (2006) 2965–2975.
[62] R. Gandikota, D. Blaauw, D. Sylvester, Modeling crosstalk in statistical static timing analysis, in: Proceedings of the International Workshop on Timing Issues, February 2008.
[63] R. Gandikota, K. Chopra, D. Blaauw, D. Sylvester, M. Becer, J. Geada, Victim alignment in crosstalk aware timing analysis, in: Proceedings of the International Conference on Computer-Aided Design, November 2007, pp. 698–704.

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